

A Theoretical Study of DBLog

Certified Virtual Cuts for a Snapshot-Equivalent Replay of Live Databases

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Abstract

DBLog is a change-data-capture (CDC) mechanism for copying a whole table or selected keys from a live source database while continuing to stream new changes. It reads the table in primary-key ordered chunks and brackets each chunk with watermarks in the source log. Rows returned by a chunk become refresh events. CDC events that are newer than a refresh prevent stale chunk values from becoming the final replay result. This approach allows backfills that can run at any time, and not only initially, while source writes and live change capture continue. It was introduced in the 2020 DBLog paper [1] and is now used in Debezium [2] and Apache Flink CDC [3].

Because chunks are read at different source-log positions, their rows do not form one physical snapshot. The 2020 paper described the mechanism operationally but left the state reconstructed by its replay implicit. We now formalize that state as a **virtual cut**: replaying a finite prefix produces the same per-key state as the source at a chosen frontier on a chosen scope. A **certified virtual cut** packages this equality in a certificate and evidence bundle carrying the source observations needed to check it. This achieves a snapshot-equivalent replay without taking a single physical snapshot. The equivalence is in the replay outcome at the chosen frontier and scope, not in a source timestamp shared by all chunk rows.

We prove that every wellformed DBLog run produces a virtual cut and that, under faithful source observation and the stated checker assumptions, an accepted certificate witnesses one. A whole-table result follows under additional coverage conditions. We also prove that a cut advances to later frontiers by appending the source CDC events for keys in scope committed in between, and that it can be restricted to smaller scopes. The results do not establish exactly-once delivery, destination-state convergence, or correctness of any named deployment. All proofs are machine-checked in Isabelle/HOL.

Keywords. databases, distributed systems, online backfill, change-data-capture, formal verification, Isabelle/HOL.

1 Introduction

DBLog was introduced in the Netflix Tech Blog in 2019 [4] and described in the 2020 DBLog paper [1]. It is a change-data-capture (CDC) mechanism for backfilling a downstream system from a live source database while continuing to stream

new changes. DBLog reads the table in primary-key chunks. A low and a high watermark place each chunk read in source-log order. Rows returned by the chunk become refresh events, while ordinary CDC events preserve changes that occur during the backfill and replace stale refresh values. Variants of this mechanism are now used by Debezium [2] and Apache Flink CDC [3], where the backfill phase is called an *incremental snapshot*.

This work is a theoretical follow-up. It assumes basic familiarity with the DBLog mechanism and does not repeat its full engineering design. Readers new to DBLog should begin with the 2020 paper or the blog post. Section 2 provides the operational detail needed for the formal development.

The mechanism was designed to allow backfills at any time, not only initially, without stalling the flow of change events and while minimizing the impact on the source database. A backfill can run during normal operation, cover an entire table or selected keys, and pause or resume between chunks. This lets operators control the scope and pace of the work without stalling live change capture. A conventional dump or blocking snapshot may be a better option if completing the backfill as quickly as possible has priority.

Reading the table in chunks while source writes continue creates a correctness question. Each chunk is read at its own source-log coordinate, and writes may commit between chunk reads. Some rows may therefore be stale by the time the backfill reaches the frontier it intends to reproduce. No physical snapshot describes all rows observed by the chunk reads. We study whether the events collected by DBLog can nevertheless be replayed to the same per-key state as the source at a chosen frontier on a chosen key scope. We also study how that equality changes when the source commits later events.

The 2020 publications explained operationally why watermark bracketing and stale-row suppression preserve the latest state. They did not state the reconstructed state as a formal object or provide a machine-checked proof. We supply that formalization.

The central object is a **virtual cut**. A virtual cut is an equality between two outcomes. On one side is the state obtained by applying a finite DBLog replay prefix. On the other side is the source state at a chosen frontier. The equality is restricted to the claimed key scope. A **certified virtual cut** packages this claim in a certificate whose evidence identifies the source observations, chunk reads, frontier, scope, and replay prefix used to justify it. The word *virtual* is important

because the equality concerns the resulting state. It does not imply that all chunk rows were read at one source timestamp.

The proof is organized in five layers. Layers 0 and 1 define source histories, replay events, abstract DBLog runs, and the conditions for a wellformed run. Layer 2 proves that every wellformed run produces a virtual cut at its frontier and scope. Layer 3 connects this run-level result to an accepted certificate under faithful source observation and the stated checker assumptions. Layer 4 derives whole-table equality when the certificate also covers the anchor domain, every key touched after the anchor, the required frontier order, and every key written by the replay. The watermark mechanics enter as premises, including the single-coordinate attribution of each chunk read described in Section 3.2. They are not derived by the abstract model. Figure 1 summarizes these dependencies. Every theorem is machine-checked in the companion Isabelle/HOL development, archived at <https://doi.org/10.5281/zenodo.21732790>.

The initial certified cut is not the end of DBLog operation. Once the chunk loop completes, DBLog continues consuming ordinary CDC events. We prove that appending exactly the source events for keys in scope between frontiers f and f' advances a cut at f to a cut at f' . We also prove that a cut on one scope remains valid when restricted to a smaller scope. Disjoint composition, concrete verification of extended certificates, and normalization of overlapping or retried chunks remain future work. Applied destination-state correctness also remains outside the current result. It requires a separate sink model with ordered and idempotent application, fencing, and visible apply-barrier evidence.

The theorems assume an ordered source log retained through the claimed frontier, valid coordinates for chunk observations, complete scope coverage, and faithful preservation of required source events and their order. Certificate acceptance checks internal consistency, not whether a connector reported the real source history. The results are source-side and scoped unless the Layer 4 conditions hold. They do not establish exactly-once delivery, destination state, verifier correctness, or correctness of a named deployment.

Section 2 develops a running example and Section 3 reviews the mechanism used by the model. Section 4 summarizes the claims and limitations. Sections 5 through 10 define the formal objects and prove the core results. Section 11 proves continuation and restriction. Section 12 separates source-side replay correctness from destination-state convergence. Sections 13 through 15 cover related work, future extensions, and verification methodology.

We use five status labels throughout (Table 1):

Each theorem states its premises directly. These labels distinguish the machine-checked result from conditions supplied by a deployment or its source connector.

2 Bootstrapping and Advancing a Virtual Cut: A Running Example

We begin with a concrete backfill before introducing the formal model. The example shows what DBLog reads while the source accepts concurrent writes, how those observations form a replay, and why that replay can match the source without being a physical snapshot. It also shows how the matching point advances when the source commits another event.

We use this example throughout the paper. This section presents the mechanism operationally, and Section 5 introduces the corresponding definitions.

2.1 Setup

A source database carries a single table:

```
accounts(account_id, balance)
```

with two live rows at the moment the example starts:

```
account_id = 100 balance = 5000 (Alice's account, $50)
account_id = 200 balance = 10000 (Bob's account, $100)
```

A third user, Carol, has not yet opened an account. Therefore, account_id 300 does not exist in the table.

During the backfill, application writes continue while DBLog reads chunks and watches the source log. We construct a replay whose per-account outcome matches the source at a chosen frontier, although the chunk reads occur at different source-log coordinates.

2.2 What the Source Log Records During the Backfill Window

While DBLog is working, four committed writes land in the source log in this order:

```
c1 Update 100 → 3000 (Alice's balance changes to $30)
c2 Insert 300 → 2500 (Carol opens an account at $25)
c3 Update 200 → 12000 (Bob's balance changes to $120)
c4 Update 300 → 3500 (Carol's balance changes to $35)
```

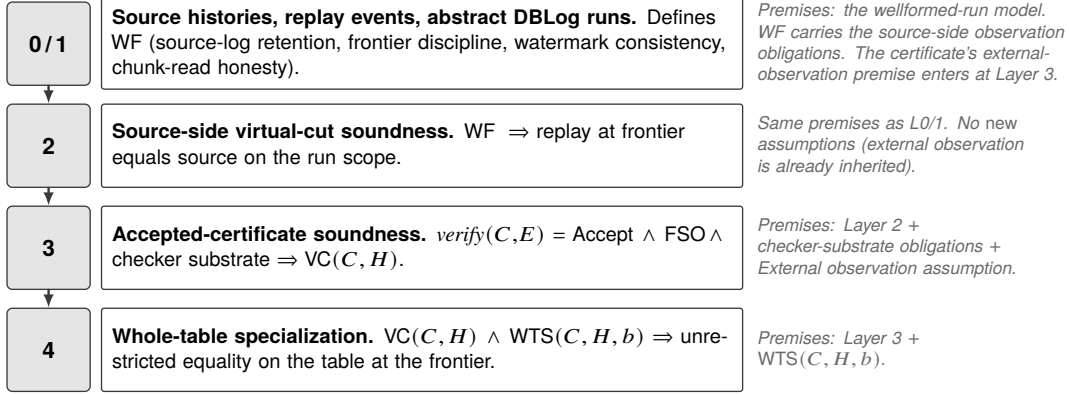
Each c_i is a source-log coordinate, a position in the source database's commit order. We will not need anything more precise than “ c_1 happened before c_2 , which happened before c_3 , which happened before c_4 .”

By the time c_4 commits, the source state on these three accounts is:

```
account 100 balance = 3000 (after the c1 update)
account 200 balance = 12000 (after the c3 update)
account 300 balance = 3500 (after the c4 update)
```

That is the per-account state we want the assembled replay to reconstruct.

The theorem ladder: source-side core (Layers 0–4)



Post-core extension map: independent extensions, *not* ladder rungs

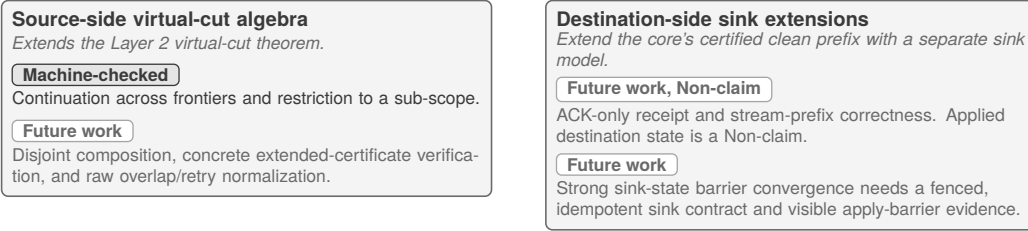


Figure 1: Theorem ladder and extensions. Layers 0–4 form the source-side core, with each layer using the premises of the layers above it. Continuation and scope restriction extend the Layer 2 virtual-cut result. Disjoint composition, extended-certificate verification, and overlap/retry normalization remain Future work. Destination-state results require a separate sink model. A cut obtained from an accepted certificate retains the certificate assumptions and faithful-source observation premise.

2.3 What the Chunks Read

DBLog reads the table while source writes continue. It splits the table into two **chunks** by primary key range:

- **Chunk A** covers $\text{account_id} \in \{100, 200\}$.
- **Chunk B** covers $\text{account_id} \in \{300\}$.

The two chunk ranges are disjoint, and together they cover every account the backfill claims. How the key space is split into chunks is a parameter chosen by the deployment, not something DBLog infers. A chunk may also cover keys that do not exist in the table yet: chunk B covers account 300 before Carol has opened it.

Each chunk is read at its own moment in source-log time. That a multi-row chunk read counts as a read at one such coordinate is itself a deployment obligation, made precise in Section 3.2.

Chunk A runs at coordinate c_1 , so its read sees the state immediately after c_1 's update committed:

```
chunk A read at c1:
account 100 balance = 3000 (already updated by c1)
account 200 balance = 10000 (no write to 200 yet)
```

Chunk B runs later, at coordinate c_3 , so its read sees Carol's

row after Carol's c_2 insert but before Carol's c_4 update:

```
chunk B read at c3:
account 300 balance = 2500 (Carol's initial balance from c2)
```

The two chunks run at different source-log moments. No single physical snapshot of the table has been taken. Chunk A saw Bob at \$100. By the time chunk B was reading, Bob's balance had already moved to \$120.

Both chunk reads here find their rows present, but a chunk read can equally record a key as *absent*: had chunk B run before Carol's c_2 insert, it would have observed account 300 not yet in the table, and that observed absence is replayed like any observed value.

2.4 The Chosen Point of Equivalence

The backfill targets a specific source-log moment for the assembled replay to match. We pick **frontier** $f = c_4$: the source-log position immediately after the last write in the sequence. The **scope** of the backfill is the set of accounts we are claiming: $K = \{100, 200, 300\}$ (Alice, Bob, Carol).

Deterministic replay of the gathered events equals the source state at f on K , although no chunk is read at f , the three

Table 1: Reader-facing status vocabulary used throughout the paper.

Status	Meaning in this paper
Machine-checked	The claim is proved in the companion Isabelle/HOL development, under premises stated where the claim appears. The paper gives the proof idea or proof explanation in the body.
Deployment obligation	The operator or implementation must establish this condition in a deployment.
External observation assumption	The source-side observation is assumed faithful. The model cannot prove it from the certificate alone.
Future work	A named extension or theorem path that this paper does not claim.
Non-claim	A property that the paper does not prove.

accounts are never read together, and source writes continue throughout.

2.5 How DBLog Assembles the Replay

DBLog treats every observed source-log event as an ordinary **CDC event** (the captured record of one committed source write: an insert, update, or delete at its source-log coordinate), every chunk-read row as a **refresh event**, and concatenates them in source-log order. Such an event is a *refresh* because it re-states a key’s current stored value, read directly from the table, rather than reporting an incremental change as a CDC event does. For our example the assembled replay sequence is:

1. CDC at c_1 : Update account 100 $\rightarrow$ 3000
2. Refresh from chunk A at c_1 : account 100 observed as 3000
3. Refresh from chunk A at c_1 : account 200 observed as 10000
4. CDC at c_2 : Insert account 300 $\rightarrow$ 2500
5. CDC at c_3 : Update account 200 $\rightarrow$ 12000
6. Refresh from chunk B at c_3 : account 300 observed as 2500
7. CDC at c_4 : Update account 300 $\rightarrow$ 3500

Both refreshes from chunk A sit at c_1 . The refresh from chunk B sits at c_3 . CDC events sit at the source-log coordinate they were committed at. Where a CDC event and a chunk refresh share a coordinate (as item 1 and items 2–3 all sit at c_1), the CDC event is placed first, because a chunk read at a coordinate already reflects every write committed at that coordinate. Figure 2 places these chunk reads and CDC events on the source-log timeline.

This is the **raw replay** form: it shows the chunk refreshes DBLog gathered even when a later CDC event for the same key supersedes them. The normalized clean prefix may drop dominated refreshes such as item 3 for account 200 and item 6 for account 300, but *apply* produces the same final per-key state from the raw and normalized forms. The raw sequence shows what DBLog observed. The normalized sequence makes stale-refresh suppression explicit and is useful in the formal model.

To reconstruct per-account state, we play the seven raw items left to right, each one writing the per-account map. When two items concern the same account, the later one wins for that account’s value.

Walking through:

- Item 1 sets account 100 to 3000.

- Item 2, chunk A’s refresh of account 100, confirms 100 = 3000. No change.
- Item 3, chunk A’s refresh of account 200, sets account 200 to 10000.
- Item 4 introduces account 300 at 2500.
- Item 5 sets account 200 to 12000. This overrides chunk A’s earlier observation of account 200: chunk A saw \$100 because it read at c_1 , but the c_3 log event is newer and authoritative.
- Item 6, chunk B’s refresh of account 300, confirms 300 = 2500. No change yet.
- Item 7 sets account 300 to 3500. This overrides chunk B’s earlier observation of account 300: chunk B saw \$25 because it read at c_3 , but the c_4 log event is newer and authoritative.

The final per-account state, restricted to scope $K = \{100, 200, 300\}$:

```
account 100 balance = 3000
account 200 balance = 12000
account 300 balance = 3500
```

Compare against the source state at $f = c_4$ restricted to K :

```
account 100 balance = 3000
account 200 balance = 12000
account 300 balance = 3500
```

They match. The assembled replay reproduces the source state at f on every account in the scope K , even though chunk A and chunk B each read at a different earlier moment, and even though one of chunk A’s observations and chunk B’s only observation were both already stale by the time f was reached.

2.6 Advancing the Cut to a Later Frontier

The seven-item replay above is a virtual cut at one frontier: applying it reaches the source state at $f = c_4$ on the scope $K = \{100, 200, 300\}$. A DBLog deployment does not stop there. The chunk loop has closed: every chunk in the plan has been read once (here, chunks A and B), and no further chunk reads will occur. However, the source keeps committing writes and DBLog keeps consuming the source log, so the cut can be advanced.

Suppose one more write lands after c_4 :

DBLog does not re-read a chunk to account for this change. The single CDC event committed after c_4 and through c_5 is the *continuation segment* for that interval. Appending that segment advances the cut, provided that the CDC feed delivers it faithfully and the source log still retains the interval. Section 11 states these conditions precisely. The replay gains one item:

8. CDC at c_5 : Delete account 200

Playing item 8 after the seven-item replay makes account 200 absent. As with an insert or update, the later source-log event overrides both the earlier refresh and the c_3 update. For a delete, the resulting state is absence.

The eight-item replay is now a virtual cut at the later frontier $f' = c_5$. Its per-account state on K is account 100 = 3000, account 200 absent, account 300 = 3500, which is the source state at c_5 on K . The point of equivalence moved from c_4 to c_5 by appending the one event the source committed in between. No chunk was re-read. Had that event been an ordinary insert or update rather than a delete, it would have appended in exactly the same way. This is the running example's instance of the continuation result of Section 11: ongoing DBLog operation is a sequence of virtual cuts, each later cut the earlier one extended by the source's faithful change data.

This says nothing, by itself, about whether a destination consuming the replay has applied item 8 or reached the state in which account 200 is absent. That remains a separate sink-side question (Section 12).

2.7 What “Snapshot-Equivalent” Means Here

At no point in this example did DBLog read the entire table at one source-log position. Chunk A saw account 100 and account 200 at c_1 . Chunk B saw account 300 at c_3 . Because the two reads happened at different points in the source database's commit order, DBLog must reconcile them through the source log rather than read them as one instantaneous observation.

What DBLog *does* have is enough source-log evidence around each chunk read to know which observations are still authoritative at the chosen frontier and which have been superseded. That is the content of the assembled replay sequence above: a deterministic mix of chunk-read refreshes and CDC events that, taken together, have the same per-key replay outcome as the source state at f on the scope K .

In this paper, *snapshot-equivalent replay* means that the replay outcome equals the source state at f restricted to K . It does not mean that DBLog performed a physical snapshot read.

2.8 Scope of the Example

The example covers one three-key backfill. It establishes replay equality for that scope, but it does not establish the following:

- **It is not a physical cut.** DBLog never performed a single read covering all of $\{100, 200, 300\}$. Chunk A read accounts 100 and 200 at c_1 , and chunk B read account 300 at c_3 . The equivalence is *extensional*. Outcomes match at f on K , although DBLog did not take a snapshot.
- **It is not a whole-table claim.** The example claims only the three accounts in K . Other accounts in the source table are not addressed by this example. Whole-table correctness is a derived specialization handled later in the paper, with extra conditions attached.
- **It is not a sink-state claim.** What we have shown is the assembled replay sequence. Whether a sink that consumes this sequence converges to the same per-account state is an additional contract on the consumer, including idempotency, ordering, and fencing. The example does not establish that contract.

2.9 Required Coordination

Avoiding a physical snapshot read does not remove the need for coordination. The example relies on the following source-side conditions:

- A **source-log ordering** that is monotone and well-defined enough to compare the chunk-read coordinate of chunk A (c_1) to the CDC event for account 200 at c_3 and conclude that c_3 is later. Without a comparable ordering on commit positions, the “later wins” rule cannot be applied.
- **Chunk-read watermarks** placed in the source log around each chunk read, so DBLog can locate chunk A's read in source-log order: the bracket places its observation of account 200 before the c_3 event for account 200, so that event is recognized as strictly newer. The example treats the chunk-read coordinate as known. The source-side mechanism that makes it known is the watermark bracket, made precise in the next section.
- **Source-log retention:** the source log must still carry c_1 through c_4 when DBLog assembles the replay. If the log had truncated c_1 before chunk A was reconciled, the assembled replay would have a gap and DBLog would not be able to claim the scope K up to f .
- **Chunk-read consistency:** chunk A's two rows must both reflect the source at the single coordinate c_1 . The watermark bracket locates the read in source-log order but does not by itself establish that single-coordinate attribution. A consistent statement-level read, or per-row coordinate evidence, is what the deployment must supply. Section 3.2 makes this obligation precise.

The deployment must establish these conditions. The formal model records them as explicit predicates and assump-

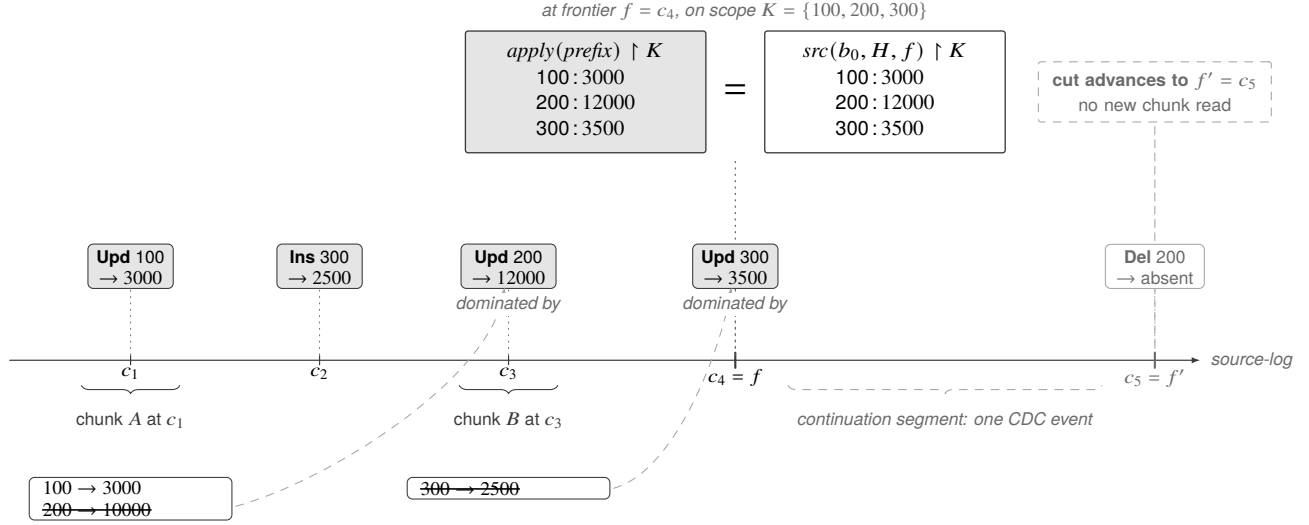


Figure 2: Two chunk reads at different source-log coordinates assemble into a replay whose state at frontier $f = c_4$ equals the source on scope $K = \{100, 200, 300\}$. Stale observations (strikethrough) are dominated by later source-log writes. No single physical read gathers every key in scope, so the cut is virtual rather than physical. That is the bootstrap cut at c_4 . The lighter elements to the right show the continuation of Section 2.6: one further source-log event (c_5 , a delete of account 200) is appended as the continuation segment for the half-open interval $(c_4, c_5]$. It begins strictly after the frontier c_4 and requires no additional chunk read. This advances the same cut to the later frontier $f' = c_5$.

tions.

2.10 Connection to the Formal Model

Section 3 defines the operational terms used above, including chunks, watermarks, refresh events, CDC events, replay prefixes, frontiers, and scopes. Sections 5 through 11 then give these terms formal definitions and prove the general results illustrated here. The seven-item replay matches the source at $f = c_4$ on K . Appending the eighth item advances the same equality to $f' = c_5$.

3 The DBLog Mechanism

The running example used two chunks and four source-log events. We now define the operational parts of the general mechanism. Figure 3 shows the example on a source-log axis, including the chunk reads, refresh windows, and merged raw replay.

3.1 Chunks and Scope

DBLog begins with a **scope**, which is the set of primary keys for which the backfill will make a claim. In the example, the scope was the three account ids $\{100, 200, 300\}$. In a larger deployment the scope might be a table, a shard, a tenant, or a selected key range.

The scope is partitioned into **chunks**. The chunks are pairwise disjoint and together cover the claimed scope. Each key in the scope therefore belongs to one responsible chunk. The deployment chooses the chunk ranges, and the model takes that plan as an input. A chunk may also cover keys that do not yet exist in the table. An observed absence within the chunk range is evidence in the same way as an observed value.

Unique ownership is required by the current model. If two chunks own the same key, or if no chunk owns a key in scope, the replay lacks one defined source for that key's refresh. Overlapping and retried chunks require a normalization step and are left as a future extension.

The model also abstracts from how the finite chunk plan is discovered. A deployment must still know when the chunk loop has reached the intended end of the claimed key scope. For a whole-table run, a common way to establish that fact is to record the table's maximum primary key, or an equivalent end-of-table observation, with enough source-side context to know which table population the bound refers to. For a selected range, shard, tenant, or other scoped run, the analogous obligation is an end-of-scope witness for that chosen scope. The deployment must provide this witness. The model separately assumes that the primary-key order and the source observation used for the bound are faithful.

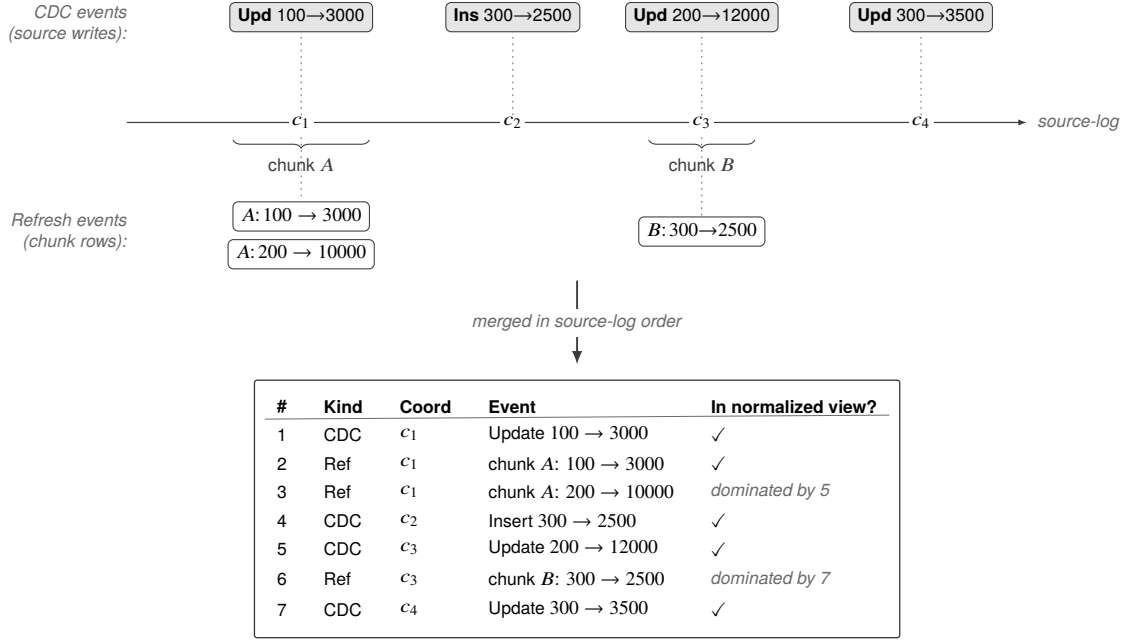


Figure 3: The DBLog mechanism in the running example. Source writes become *CDC events* above the axis. Chunk-read rows become *refresh events* (below the axis, grouped by chunk and bracketed by source-log watermarks). The two streams merge in source-log order into the *raw clean prefix* below, which is the list used by the Layer 2 theorem. A refresh row whose key is touched by a later source write is marked *dominated*. The raw clean prefix keeps it, and an optional normalized view may drop it without changing the *apply* result (§3.4).

3.2 Watermarks and Chunk-Read Coordinates

A chunk read must be located in source-log time. DBLog needs to know which source events precede the read and which events can supersede its rows. It obtains this ordering with **watermarks**. Before a chunk read, DBLog writes a *low watermark* to a dedicated table in the source. After the scan, it writes a *high watermark*. Both writes appear in the source log at definite coordinates, with the chunk SELECT between them. The baseline mechanism therefore needs write access to the source. A read-only deployment needs another source coordinate witness.

The watermark pair establishes the read’s *provenance* in source-log order. The formal model attributes the returned rows to one *chunk-read coordinate*, such as c_1 for chunk A in the running example. Every row image is treated as faithful to the source at that coordinate.

The watermark bracket alone does not establish this single-coordinate attribution. A multi-row SELECT is not instantaneous. The deployment must also provide either read semantics that make all rows consistent at one source coordinate, such as a statement-level snapshot or repeatable-read scan, or per-row coordinate evidence. Thus, “chunk A read at c_1 ” combines two facts. The watermarks bracket the read at c_1 , and the returned row images are faithful to the source at c_1 .

In the original DBLog mechanism, the interval between the two watermarks is also the window over which stale chunk rows are suppressed. A chunk row is discarded when the source log shows a write for the same key inside that window. The model represents that behavior with two simpler parts. Each returned row has a faithful chunk-read coordinate, and replay applies a frontier-wide *latest-event-wins* rule (§3.3). The two views reach the same per-key state, by the window-suppression equivalence of §3.4. A write committed inside the watermark window is itself an ordinary CDC event the replay carries, and latest-event-wins lets it override the chunk row in the same way as in-window suppression. A write after the high watermark is handled identically. Suppression in the window and later override at the frontier therefore produce the same state.

The formal run retains the low and high watermarks as evidence for the chunk-read coordinate. They do not add another state-equality condition. Section 5.5 records the required source-log ordering, retention, coordinate provenance, and row-image faithfulness. The last condition appears there as “Refresh honesty.”

DBLog avoids a single physical snapshot read. It does not avoid source-log coordination.

3.3 Refresh Events and Ordinary CDC Events

DBLog turns each chunk-read row into a **refresh event**. A refresh records that key k held a given value, or was absent, at the chunk-read coordinate. A source-log write becomes an ordinary CDC event that records an insert, update, or delete at its source-log coordinate.

Both kinds of event enter one replay vocabulary. Ordinary CDC events carry committed source writes. Refresh events carry row observations from chunk reads. Applying a replay means walking the event sequence left to right and updating the per-key state. Later events for the same key win. That behavior is what made Bob's c_3 update override chunk A's older refresh in the example.

3.4 Raw Replay and Normalized Clean Prefix

DBLog assembles a finite sequence called the *clean prefix*. We use two forms of this sequence. The theorems use the *raw* form. The *normalized* form is an optional, *apply*-equivalent presentation.

The **raw replay** is the evidence-preserving sequence: it may include a refresh even if a later CDC event for the same key dominates it. Raw replay is useful operationally because it shows exactly what DBLog observed.

The **normalized clean prefix** makes stale-refresh suppression explicit. It may omit refresh events that are dominated by newer source-log evidence, provided suppressing them does not change the final *apply* state. In the example, Bob's chunk-A refresh was dominated by Bob's later c_3 update. Carol's chunk-B refresh was dominated by Carol's later c_4 update. Carrying or suppressing those dominated refreshes gives the same final per-key state. The proof-facing bridge is state equivalence on the claimed scope: suppressing only same-key dominated refreshes preserves *apply* for every claimed key.

Window-suppression equivalence. *Retaining or suppressing a same-key refresh that a later source event dominates leaves apply unchanged on every claimed key.* This fact connects the raw and normalized prefixes. It also connects original DBLog's in-window suppression to frontier-wide latest-event-wins. In both cases, a later same-key source event dominates the refresh, so *apply* produces the same state. Section 7 uses this equivalence to state the Layer 2 theorem about the raw clean prefix while leaving normalization an optional presentation.

The theorem later relies on these facts:

- no required CDC event is missing.
- no stale refresh is allowed to resurrect an older value past a newer event.
- suppressing a refresh is justified only when newer source-log evidence dominates it for the same key.

3.5 Frontier and Clean Prefix

The **frontier** is the source-log coordinate at which a cut is claimed (concretely, a MySQL executed global-transaction-identifier (GTID) set, a PostgreSQL log sequence number (LSN), or a CockroachDB resolved timestamp). A certificate claims equality at a particular frontier, not at an unspecified eventual state. The clean prefix contains the replay evidence needed through that frontier for the claimed scope.

Operationally, a DBLog certificate is useful only when the operator can say:

for this scope K and frontier f , here is the finite replay prefix whose application equals the source state at f on K .

The rest of the paper makes that sentence precise. Section 5 defines the source and replay models. Section 6 names the equality as a virtual cut. Section 7 proves the source-side theorem for wellformed runs. Sections 8 and 9 move from runs to accepted certificates.

The clean prefix is finite but not necessarily small. It carries a refresh for each key in the claimed scope and every in-scope CDC event through the frontier, so it scales with the scope and the change volume over the certified interval. For a large table backfilled over a busy window, the clean prefix is correspondingly large. The normalized clean prefix trims refreshes that newer evidence dominates. The raw form, on which the theorems are stated, does not.

3.6 After the Chunk Loop: Advancing the Frontier

The backfill chunk loop runs until the claimed key scope is covered and then closes. DBLog does not stop at the first frontier. Once the chunk loop has established a virtual cut at a frontier f , DBLog continues consuming ordinary CDC events as the source commits them. The source-side result does not need a fresh chunk read for every later frontier. If the source log faithfully supplies the change data the source committed after f and through a later frontier f' , appending that segment to the clean prefix advances the replay-equivalence claim from f to f' .

This is the operational reading of the continuation result proved in Section 11: the backfill bootstraps the first virtual cut, and faithful CDC continuation advances it. The chunk reads happen once. A later cut is obtained by appending faithful source-log change data. This step relies on ordered source-log coordinates and retention of the events between f and f' . Section 11 states the result and its premises.

Practitioner takeaway. *DBLog combines a finite chunk plan, source-log watermarks, chunk refreshes, and ordinary CDC events into a replay at a named frontier. The deployment must provide an ordered and retained source log, faithful chunk reads, and complete coverage of the claimed scope.*

4 DBLog Claims and Limitations

DBLog combines source-log events with chunk-read observations. A live backfill establishes an initial virtual cut, and later CDC events can advance that cut to new source frontiers. The result is a source-side replay claim on a named scope and frontier. It does not remove the need for source coordination or establish the state of a destination.

4.1 What DBLog Provides

The protocol combines the following inputs:

- a claimed key scope.
- a certified frontier in source-log order.
- chunk reads whose observations become refresh events.
- ordinary CDC events from the source log.
- source-log and chunk-read evidence sufficient to justify the clean prefix.
- a finite replay prefix whose *apply* state can be compared with the source.

Replaying the clean prefix reaches the same per-key state as the source at the frontier on the claimed scope. Once this equality has been established, appending the source changes committed after that frontier and through a later frontier advances the cut. Neither step requires a physical snapshot read.

4.2 Limits of the Result

The result depends on source-log ordering, retained CDC events, chunk-read coordinates, frontier discipline, and evidence that ties these observations to the source. A deployment or source connector must establish these conditions.

The core theorem is scoped. Whole-table equality is derived only when the Layer 4 anchor-domain and no-unclaimed-replay-key conditions also hold.

The theorem does not establish exactly-once delivery or destination state. A destination may receive the certified replay without applying it in the required order or reaching the corresponding state. The source-side continuation result advances replay equality to later frontiers, but it does not show that a sink has applied those events. Section 12 states this limit.

A virtual cut also does not imply that one physical source-read timestamp existed for all chunk rows. It is an equality of outcomes.

Debezium, Apache Flink CDC, MySQL, PostgreSQL, and MariaDB provide deployment and related-work context. The theorem does not apply to any named system until its implementation and source assumptions have been checked against the formal model.

4.3 Failure Modes and Required Evidence

Table 2 lists the main failure modes, the evidence or assumption that addresses each one, and whether the condition is machine-checked or supplied by the deployment.

For the initial certificate, a checker can expose the internal evidence failures listed in the table. Layer 4 adds the conditions needed for a whole-table claim. A concrete checker for continuation certificates is not included. In all cases, the theorem assumes that the source observations are faithful to the mathematical source history.

4.4 Trust Boundary

The strongest result remains source-side. Under faithful source observation and the checker assumptions, accepted evidence yields a certified virtual cut. The result does not prove that a concrete parser is correct, that a connector reports the source faithfully, that a named product implements every assumption, or that a sink has applied the replay.

Practitioner takeaway. *A certificate supports a source-side replay claim on a chosen scope and frontier. It does not by itself establish exactly-once delivery, whole-table coverage, destination-state convergence, or correctness of a named product.*

5 Formal Model (Layer 0 and Layer 1)

Layer 0 defines source histories, source state at a frontier, replay events, replay application, and restriction to a key scope. Layer 1 defines abstract DBLog runs, including chunks, watermarks, chunk reads, CDC observations, clean prefixes, and wellformedness conditions.

These layers do not include certificates, verifiers, destination ACKs, or whole-table conditions. Layer 2 proves a theorem about wellformed runs. Layer 3 then states when an accepted certificate witnesses such a run.

Table 3 fixes the symbols the formal sections use. A predicate (short sans-serif tag like WF, FSO, Mat, Coh, VC, WTS) is a relation that holds or fails on its arguments. An accessor function (italic word like *apply*, *src*, *scope*, *frontier*, *prefix*) returns a value.

5.1 Source Histories and Frontiers

A source database is modeled as a base state b_0 together with a finite history H of committed source events. A source event is one of

Insert $k\ v$, Update $k\ v$, Delete k .

Each event carries a source coordinate: a commit position, log sequence number, monotone source-side timestamp, or

Table 2: Failure modes, required evidence or assumptions, and their status.

Failure mode	What goes wrong	Evidence or assumption boundary	Status
Missing CDC event	A source write in the claimed interval is absent from the replay evidence.	Source-log retention and CDC coverage through the certified frontier.	Deployment obligation / External observation assumption
Missing continuation segment	A later frontier is claimed without the faithful source-log change data the source committed between the certified frontier and that later frontier.	Source-log retention for the continuation interval and faithful CDC coverage.	Deployment obligation / External observation assumption
Bad watermark or coordinate	A chunk row is treated as if it was read at the wrong source-log position.	Watermark or coordinate faithfulness for the chunk read.	Deployment obligation / External observation assumption
Stale refresh survives	An older chunk-read value overrides a newer source-log event for the same key.	Clean-prefix dominance and latest-event ordering.	Machine-checked
Wrong frontier	The replay is certified against a frontier different from the source interval it actually covers.	Frontier consistency in the certificate and evidence.	Machine-checked
Scope ownership mismatch	A claimed key is not owned by exactly one responsible chunk, or the claimed scope does not match the chunk plan.	Chunk ownership and claim-scope checks.	Machine-checked
Incomplete chunk-plan termination	The chunk loop stops before the intended end of the key range, so a key in the intended operational scope may be absent from the finite plan. For a whole-table claim, the anchor domain is not shown to be covered.	Chunk-plan construction, maximum-key or end-of-scope evidence, primary-key ordering, and whole-table claim-scope coverage.	Deployment obligation / External observation assumption
Whole-table replay-key mismatch	The whole-table specialization is requested while replay evidence materializes keys outside the claim scope.	Layer 4 no-unclaimed-replay-key side condition, which is not part of scoped virtual-cut equality.	Machine-checked
Unfaithful observation	The connector, source log, primary-key interpretation, or chunk-read result is not faithful to the real database.	Not internally provable from the certificate alone.	External observation assumption

equivalent order-bearing token. We write $c \leq c'$ when coordinate c is no later than c' . A source history is wellformed when its events are ordered by that coordinate, the base coordinate sits before every event, and same-coordinate events have a deterministic source-position order inside the history.

The source state at a frontier is written $src(b_0, H, f)$. For each key k , this state looks for the latest source event for k at or before frontier f . An insert or update makes the key present with the written value. A delete makes it absent. If there is no event, the key keeps its base state $b_0(k)$.

The coordinate convention is inclusive. A chunk read recorded at coordinate c observes $src(b_0, H, c)$, including source events at coordinate c . A chunk read is *attributed* to the latest coordinate whose committed effects its row images already reflect. §3.2 sets out what a deployment

must supply for that attribution to be sound. The watermark bracket locates the read, while chunk-read semantics or per-row coordinate evidence make the returned row images faithful to one coordinate. Canonical replay therefore places same-coordinate CDC evidence before the refresh that records the post- c chunk-read state. Later source-coordinate events still appear after that refresh.

This is the source-side reference state that every virtual-cut theorem compares against.

Table 3: Notation used in the formal model and theorem statements. Predicates are rendered in sans-serif. Accessor functions are italic.

Symbol	Kind	Read as
b_0, H, f	variables	base state, source history, frontier coordinate.
b, f'	variables	whole-table anchor source coordinate ($b \leq f$) and later source frontier as the continuation target ($f \leq f'$).
R, C, E	variables	abstract DBLog run, certificate, evidence bundle (§8.2).
σ, δ, K	variables	replay sequence, continuation segment (CDC replay-event list), key scope.
$src(b_0, H, f)$	accessor	source state at frontier f .
$apply(\sigma)$	accessor	replay-apply of sequence σ .
$scope(\cdot)$	accessor	claimed key scope (of run or certificate).
$frontier(\cdot)$	accessor	chosen source-log frontier.
$prefix(\cdot)$	accessor	clean replay prefix.
$m \upharpoonright K$	operator	state m restricted to keys in K .
$verify(C, E)$	accessor	verifier result, one of <i>Accept</i> , <i>Reject</i> , <i>Unsupported</i> .
raw	variable	raw source-observation carrier (External observation assumption).
$anchorDom(H, b)$	accessor	keys present at the anchor boundary b .
$touched(H, b, f)$	accessor	keys with source events in $(b, f]$.
$tblScope(H, b, f)$	accessor	union of the two above.
$keys(\sigma)$	accessor	keys written by replay sequence σ .
$chunks(R)$	accessor	run’s chunk plan.
$srcHistory(R)$	accessor	source history the run is about.
$cdcEvents(R)$	accessor	CDC events the run consumed.
$WF(b_0, R, H)$	predicate	wellformed DBLog run (Layer 1).
$VCS(\sigma, K, f, H)$	predicate	state-level virtual cut (Section 6). The base state b_0 is implicit in this four-argument shorthand and made explicit in the unfolded form in Section 6.
$VC(C, H)$	predicate	certificate-side virtual cut.
$ContSeg(H, K, f, f', \delta)$	predicate	δ is the continuation segment for $(f, f']$ on K (Section 11).
$Mat(C, E, R)$	predicate	(C, E) materializes abstract run R .
$Coh(C, E, R)$	predicate	(C, E) is coherent with run R (Section 8.4).
$FSO(E, b_0, H, raw)$	assumption	faithful source observation.
$WTS(C, H, b)$	predicate	whole-table claim scope (Layer 4).

5.2 Replay Events and Apply

DBLog does not replay source events alone. It replays two kinds of replay event:

$Cdc\ c\ e$ source-log event e at coord. c ,
 $Refresh\ k\ m\ c$ chunk-read of k at coord. c .

In a refresh, $m = \text{Some } v$ means the chunk read observed value v . The value $m = \text{None}$ means that the row was absent. Absence is part of the state. Replaying a refresh with *None* deletes the key from the replay state.

The replay function is $apply(\sigma)$, where σ is a finite replay sequence. $apply$ walks the sequence from left to right, updating a per-key map. Later events for the same key win. $apply$ starts from the all-absent per-key map. Formally, $apply([\])(k) = \text{None}$ for every key k . Thus the running example’s later update for Bob overrides chunk A’s older refresh, and a later delete would override both an earlier refresh and an earlier update.

State restriction is written $m \upharpoonright K$ in prose formulas below. It keeps only the entries whose keys lie in K , so $(m \upharpoonright K)(k) =$

$m(k)$ when $k \in K$, and *None* otherwise. The displayed formulas use the $\upharpoonright$ symbol throughout.

5.3 Abstract DBLog Runs

A DBLog run R is modeled abstractly. The model observes it only through a small set of accessors:

$scope(R)$ claimed key scope,
 $frontier(R)$ chosen source frontier,
 $prefix(R)$ replay sequence of the run,
 $chunks(R)$ run’s chunk plan,
 $srcHistory(R)$ source history the run is about,
 $cdcEvents(R)$ observed CDC events consumed.

Each chunk has a domain of keys, a responsible-chunk relation from keys to chunks, a lower and upper watermark, and a chunk-read coordinate. The model uses a strict canonical partition. Each key in $scope(R)$ belongs to exactly one responsible chunk, and every responsible chunk has a finite domain. The model also takes the chunk plan itself to be

Table 4: Wellformed-DBLog-run obligations: clauses of $WF(b_0, R, H)$ in reader-facing terms, with status.

Obligation	Meaning	Status
Source binding	R is about source history H , and H is wellformed.	Deployment obligation
Chunk partition	The chunks form a finite strict partition of $scope(R)$.	Deployment obligation
CDC coverage	Every relevant source event after the responsible chunk read and through the frontier is observed.	External observation assumption
CDC faithfulness	Observed CDC events are real source events, in the right order, with same-key same-coordinate multiplicity preserved.	External observation assumption
Chunk evidence	Chunk read coordinates are bracketed by watermarks, and chunk-read observations produce the corresponding value-or-absence refreshes for claimed keys.	Deployment obligation
Read-before-frontier	Every chunk read occurs no later than the run frontier.	Deployment obligation
Refresh honesty	A chunk-read value or absence matches $src(b_0, H, \text{read-coord.})$ for that key.	External observation assumption / Deployment obligation

finite, so every run has a finite set of chunks. This models a backfill with a bounded chunk loop over a finite plan.

That strict partition is not a claim that real systems never retry or overlap chunk reads. It is the canonical run selected for the theorem. Raw overlap/retry evidence must be normalized into such a canonical run, or handled by a future extension.

5.4 Clean Prefix

The clean prefix of a run is the finite replay sequence produced from:

- chunk-read refreshes for the responsible chunks.
- observed CDC events for in-scope keys through the frontier.
- the *clean-prefix ordering*. Events are sorted by source coordinate, with a CDC event placed before a chunk refresh at a shared coordinate, and same-coordinate CDC events kept in source-log position order.
- suppression, or harmless retention, of refreshes dominated by newer same-key source-log evidence.

The theorem does not rely on an unstated “good replay” primitive. The clean prefix is constrained by the run’s chunk plan, CDC observations, frontier, and scope. If a needed CDC event is missing, if a future event is replayed, if a refresh appears without a responsible chunk observation, or if same-coordinate CDC order is scrambled, the run is not wellformed for the theorem.

The main invariant is per-key. For each key in scope, the latest replay event in the clean prefix is either the latest faithful source event through the frontier, or it is a refresh whose value or absence at its read coordinate remains unchanged until the frontier. Layer 2 derives its result from this invariant.

The mechanization has one additional premise. The canonical clean prefix enumerates each chunk domain in sorted key order, so the run- and certificate-layer theorems assume a linear order on the key type. This is no extra demand, since chunks are primary-key ranges and ranges already presuppose that order. The state-level continuation and restriction

theorems of Section 11 carry no such constraint.

5.5 Wellformed Runs

We write $WF(b_0, R, H)$ for the wellformed DBLog run predicate. It collects the source-side conditions used by the proof. Table 4 lists these conditions in reader-facing terms.

The split between Deployment obligation and External observation assumption is intentional. The model can state what a wellformed run must satisfy. It cannot prove, from inside the run, that an external database connector reported every source event faithfully or that a chunk read was faithful to the real source.

Refresh honesty carries both labels because it combines two facts. A returned row image must report the source value or absence faithfully, which is an External observation assumption. The chunk-read semantics pin those images to a single coordinate (§3.2), which is a Deployment obligation.

The predicate has one further conjunct, clean-prefix closure, omitted from the table because it follows from the construction in Section 5.4 rather than from a deployment or external observation. The proof may therefore use it without adding a deployment obligation.

The lower and upper watermarks introduced above are source-log coordinates, not primary-key-range bounds. Accordingly, the Chunk partition row is a statement about the finite canonical partition consumed by the theorem. It is not a proof that a concrete chunk-selection loop stopped in the right place, and Layer 1 contains no maximum-key accessor or chunk-loop termination algorithm. A maximum-primary-key query, an empty-next-chunk observation, a predeclared range bound, or another end-of-scope witness may justify the finite partition operationally. Supplying the resulting partition, including evidence that the loop reached the intended end of the claimed key scope, is a Deployment obligation. The theorem consumes the partition, not the plan-construction algorithm.

5.6 Layer 0 and Layer 1 Lemmas

The source and replay model supplies the elementary facts used by the main proof:

- *apply* is latest-event-wins per key.
- Restricted states are equal exactly when they agree on every key in the scope.
- *src* is characterized by the latest source event at or before the frontier.
- A replay-side Cdc event has the same per-key effect as its source event.
- A refresh has the source-state value at its chunk-read coordinate when the chunk-read honesty premise holds.
- The source state at a later frontier is the source state at an earlier frontier updated by exactly the source events strictly between them. This is immediate from the preceding characterization, since the source events at or before f' are those at or before f together with those in $(f, f']$. This interval characterization is what the continuation result of Section 11 builds on.

The run layer then contributes the bridge lemma, displayed here as a preview of the Layer 2 theorem stated and explained in Section 7:

$$\text{WF}(b_0, R, H) \implies \text{apply}(\text{prefix}(R)) \upharpoonright \text{scope}(R) \\ = \text{src}(b_0, H, \text{frontier}(R)) \upharpoonright \text{scope}(R).$$

Section 6 first names the equality itself.

Practitioner takeaway. *The wellformed-run conditions require a bound source history, a finite chunk partition, watermark-bracketed chunk reads, and complete and faithful CDC evidence. The following theorems apply to runs that meet these conditions.*

6 Virtual Cuts

A virtual cut states that replay reaches the same per-key state as the source at a chosen frontier on a chosen scope. It is an equality of outcomes, not a physical read cut.

We write $\text{VCS}(\sigma, K, f, H)$ for the state-level form, with the base state b_0 left implicit in this four-argument shorthand:

$$\text{VCS}(\sigma, K, f, H) \iff \text{apply}(\sigma) \upharpoonright K = \text{src}(b_0, H, f) \upharpoonright K.$$

Here σ is a replay sequence, K is a key scope, f is a frontier, and H is the source history. The base state b_0 is fixed by context. When needed, the formulas write it explicitly.

The certificate-side form $\text{VC}(C, H)$ specializes the same equality to the accessors of a certificate:

$$\text{VC}(C, H) \iff \text{VCS}(\text{prefix}(C), \text{scope}(C), \text{frontier}(C), H).$$

Expanded:

$$\text{apply}(\text{prefix}(C)) \upharpoonright \text{scope}(C) \\ = \text{src}(b_0, H, \text{frontier}(C)) \upharpoonright \text{scope}(C).$$

The definition is extensional. It says that two per-key states are equal on a scope at a frontier. It does not say that every chunk row was read at that frontier, that one physical source snapshot existed, that the scope is the whole table, or that any destination has applied the replay. Figure 4 contrasts a physical cut with the virtual cut this paper proves.

This is the exact shape illustrated by the running example. There, the replay sequence has seven items, the scope is $\{100, 200, 300\}$, the frontier is c_4 , and *apply* produces the same balances as *src* at c_4 on those three keys. Chunk A and chunk B were read at different coordinates. The virtual cut does not hide that fact. It says the replay outcome is snapshot-equivalent on the claimed keys.

The state-level form is useful because Layer 2 proves the equality for run accessors:

$$\text{VCS}(\text{prefix}(R), \text{scope}(R), \text{frontier}(R), H).$$

Layer 3 then proves that an accepted certificate, under faithful source observation, witnesses a run whose accessors agree with the certificate. That is how the run-side theorem becomes the certificate-side theorem.

7 Source-Side Virtual-Cut Soundness (Layer 2)

Layer 2 proves the source-side theorem:

$$\text{WF}(b_0, R, H) \implies \text{apply}(\text{prefix}(R)) \upharpoonright \text{scope}(R) \\ = \text{src}(b_0, H, \text{frontier}(R)) \upharpoonright \text{scope}(R).$$

Reader-facing theorem summary:

- **Status:** Machine-checked under the displayed premises.
- **Claim:** the clean prefix of a wellformed DBLog run reaches the source state at the run frontier on the run scope.
- **Premises:** the wellformed-run obligations in Section 5.5, including chunk partition, CDC coverage and faithfulness, chunk-read honesty, and read-before-frontier discipline.
- **Boundary:** this is a source-side run theorem, not yet a certificate, whole-table, or sink-state theorem.

Equivalently, every wellformed DBLog run satisfies the virtual-cut-state equality at the run's frontier on the run's scope. The theorem is stated about the run's raw clean prefix. By the window-suppression equivalence of §3.4, the normalized clean prefix yields the same *apply* state and therefore the same equality.

The theorem assumes the wellformed-run predicate of Section 5.5, including chunk partition, frontier discipline, CDC coverage and faithfulness, and chunk-read honesty. Each

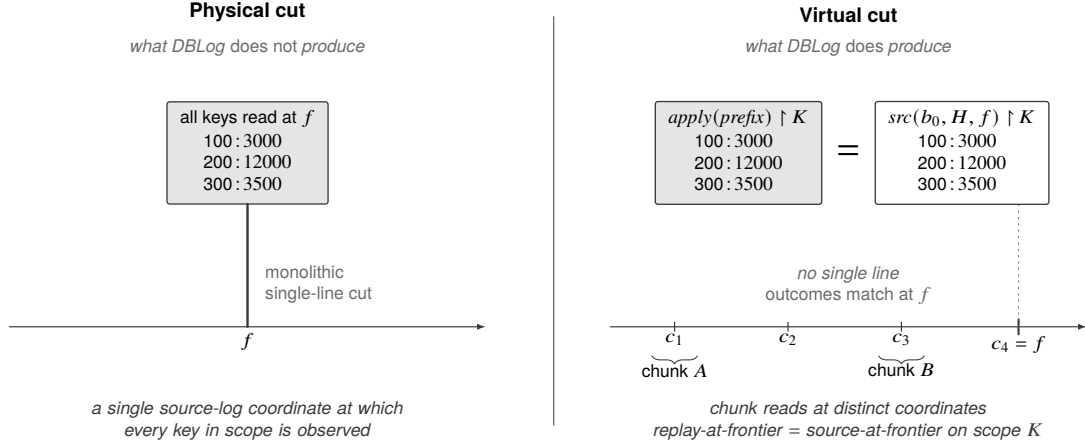


Figure 4: Physical cut vs. virtual cut. *Left*: a physical cut requires a single source-log coordinate at which every key in scope is observed, shown as a single vertical line. DBLog does *not* produce one. The values shown are the counterfactual outcome on K . *Right*: a virtual cut requires replay at the frontier to equal the source at the frontier on the chosen scope. The chunk reads (as in Figure 2) sit at distinct earlier coordinates, no single line gathers them, and the equality lives in outcome space rather than time space.

is stated explicitly because omitting it permits a real counterexample (Table 5). The predicate’s remaining conjunct, clean-prefix closure, is derived by definition. It holds for every run by the clean-prefix construction, so the proof may use it but no deployment need establish it separately.

7.1 Proof Idea

The proof is a per-key latest-event argument.

First, restricted-state equality reduces the theorem to point-wise equality. It is enough to show that for every key k in $\text{scope}(R)$:

$$\text{apply}(\text{prefix}(R))(k) = \text{src}(b_0, H, \text{frontier}(R))(k).$$

Fix such a key k . Since the run is wellformed, k has a unique responsible chunk, and the chunk-read evidence produces a value-or-absence refresh for k that belongs to $\text{prefix}(R)$, including $\text{Refresh } k \text{ None } c$ when the row is absent at the chunk-read coordinate. Therefore the “no replay event for k ” case cannot occur for an in-scope key. The clean-prefix-of-latest correspondence says that every in-scope k has a latest event in $\text{prefix}(R)$ and that this event is either a CDC event or the responsible chunk’s refresh. The proof therefore has two cases.

7.2 Case 1: The Latest Replay Event Is CDC

Suppose the latest replay event for k in $\text{prefix}(R)$ is a CDC event $\text{Cdc } c \ e$. The clean-prefix coherence obligations say this replay event comes from an observed CDC pair. CDC faithfulness says that observed pair is a real source event in H .

Because the event lies in the clean prefix, its source coordinate c is within the run’s frontier, i.e. $c \leq \text{frontier}(R)$. Order preservation and same-coordinate multiplicity say the replay did not silently reverse or drop competing source events for the same key and coordinate.

Now assume, for contradiction, that the source history has a later event for k at or before the frontier. CDC coverage would put that later event into the observed CDC list. Clean-prefix closure would put it into the replay. The clean-prefix ordering would place it after $\text{Cdc } c \ e$, contradicting the choice of $\text{Cdc } c \ e$ as the latest replay event for k .

So the latest replay CDC event corresponds to the latest source event for k through the frontier. apply and src therefore make the same update at k .

7.3 Case 2: The Latest Replay Event Is a Refresh

Suppose the latest replay event for k is $\text{Refresh } k \ m \ c_{\text{read}}$. The responsible-chunk obligations identify the chunk that produced it, and refresh honesty says:

$$m = \text{src}(b_0, H, c_{\text{read}})(k).$$

If there were a later source event for k in $(c_{\text{read}}, \text{frontier}(R)]$, CDC coverage and clean-prefix closure would place a corresponding Cdc event after the refresh in the clean prefix. The interval is open at c_{read} because a read at c_{read} already observes source events at c_{read} . The post-read repair obligation starts after the read coordinate and runs through the frontier. A later source event would contradict the refresh being latest for k .

Therefore the source state for k at the read coordinate is still the source state for k at the frontier. The refresh writes exactly that value or absence, so $apply(prefix(R))(k)$ equals $src(b_0, H, frontier(R))(k)$. The value case and the row-absence case use the same equation. Absence is None in the per-key state.

7.4 Why the Assumptions Are Necessary

The proof fails if the obligations are weakened.

These negative cases are the reason the theorem’s premise is the full wellformed-run predicate rather than a vague statement that DBLog “collected some chunks and some CDC.”

Practitioner takeaway. *A wellformed backfill produces a replay that reaches the source state at the chosen frontier on the chosen scope. The result does not require a long-lived table lock or a pause in source writes. The next two layers connect this run-level result to a certificate and, under additional conditions, to the whole table.*

8 Certificates, Evidence, and Verifiers

Layer 2 proves a theorem about an abstract DBLog run. If the run is wellformed, its clean replay prefix reaches the source state at its frontier on its scope. A deployment, however, does not hand a reader an abstract run. It hands over a certificate and evidence. This section defines those artifacts. Section 9 gives the corresponding theorem.

8.1 Certificate

A **certificate** is the claim object. It names:

- $scope(C)$ the keys being claimed,
- $frontier(C)$ the source-log coordinate of the claim,
- $prefix(C)$ the certified replay prefix.

Before acceptance, $prefix(C)$ is only the *claimed* prefix. Storing a replay list in a certificate does not authenticate it. Accepted evidence must materialize a wellformed run and tie the certificate accessors to that run. The certificate theorem uses the prefix only after accepted evidence materializes a coherent wellformed run that agrees with that accessor.

8.2 Evidence

An **evidence bundle** contains the source-side material the checker inspects. This includes certificate binding evidence, the chunk plan, chunk-read coordinates, chunk domains, observed row values or absences, ordinary CDC observations, and the construction evidence for the raw or normalized clean prefix.

The evidence bundle does not contain a destination-state proof. It does not contain sink ACK semantics. It does not

contain whole-table claim-scope side conditions except where those are added later for the whole-table specialization. The certificate layer is source-side.

8.3 Materialized Run

The materialization relation $Mat(C, E, R)$ connects certificate C and evidence E to the model. Together, they describe the abstract DBLog run R consumed by the Layer 2 theorem. Materialization is structural. It connects the evidence to a run carrier with a scope, frontier, clean prefix, chunk plan, chunk-read observations, and CDC observations. It does not by itself say that the run is wellformed against the source history.

Concretely, Mat relates the pair to the evidence-determined run. The materialized run’s scope, frontier, clean prefix, chunk plan, chunk-read observations, and CDC observations are fixed by what E serializes, not chosen freely by the checker. Evidence that does not describe a run shape the checker supports is reported **Unsupported** rather than carried into an accepted claim. The certificate’s own fields are tied to the materialized run by the coherence predicate below. Materialization alone establishes neither wellformedness nor faithfulness. Those properties come from the coherence predicate and the checker-substrate obligations.

Rejected and unsupported evidence need not materialize a run. For accepted evidence, the coherence predicate ensures that the checker uses the run determined by the evidence.

8.4 Certificate / Run Coherence

We write $Coh(C, E, R)$ for the certificate / run coherence predicate. It packages materialization with accessor agreement between C and R :

$$\begin{aligned} Coh(C, E, R) &\iff Mat(C, E, R) \\ &\quad \wedge scope(C) = scope(R) \\ &\quad \wedge frontier(C) = frontier(R) \\ &\quad \wedge prefix(C) = prefix(R). \end{aligned}$$

Coherence concerns the run materialized by this certificate and evidence pair. It is stronger than the existence of some run with matching accessors.

8.5 Verifier Result

The verifier result has three cases (Table 6):

Unsupported matters because silently ignoring unknown evidence would be a soundness bug. If the checker cannot interpret an optional source-side feature, the certificate is not accepted as if that feature had been checked.

Accept, **Reject**, and **Unsupported** are verifier results, not theorem-status labels. The verifier result here is an abstract

Table 5: Counterexamples obtained when selected obligations in WF are removed.

Missing obligation	Counterexample shape	Boundary protected
CDC coverage	A source update after a chunk read is missing from replay, so an old refresh wins.	DBLog needs retained source-log evidence.
Strict chunk ownership	Two chunks both claim a key, or no chunk owns it, so the refresh source is ambiguous or absent.	The theorem is scoped through a canonical chunk plan.
Read-before-frontier	A chunk read after the frontier injects a future value into the replay.	The replay is frontier-bound.
Same-coordinate order and multiplicity	Duplicate or same-coordinate source events are observed in a different order or with an occurrence missing.	Latest-event reasoning must match source order.
Refresh honesty	A chunk refresh records a value or absence that was not the source state at the read coordinate.	Chunk reads remain an External observation assumption.
Clean-prefix order	A clean-prefix event for some key sits before an earlier-coordinate same-key event, so the prefix’s latest event for that key differs from the source’s.	Per-key latest-event reasoning carries from the prefix to the source.

Table 6: Verifier result cases.

Result	Meaning
Accept	The certificate/evidence pair is internally coherent for the source-side checker fragment.
Reject	The evidence contradicts the certificate or fails a required check.
Unsupported	The evidence uses source-side vocabulary outside the checker fragment modeled here.

source-side checker result, not a claim that a serialized certificate format, parser, or production verifier implementation has been proved correct.

8.6 What Accept Means, and What It Does Not Mean

Accept means that, under faithful source observation, the evidence materializes a wellformed DBLog run whose scope, frontier, and clean prefix agree with the certificate. That is the bridge back to Layer 2.

Accept does not validate the concrete database, connector, log, primary-key interpretation, source-coordinate semantics, parser, wire format, or named product. It also does not show that a sink applied the prefix or that the claim covers the whole table. Faithful source observation remains a premise.

8.7 Faithful Source Observation

We write $\text{FSO}(E, b_0, H, \text{raw})$ for the faithful-source-observation assumption, the External observation assumption of the certificate layer. It ties the finite evidence bundle E to the mathematical source history H and base state b_0 , against a raw source-observation carrier raw . It covers the following facts:

- the source history and base state the certificate is about.
- the meaning and ordering of source-log coordinates.
- CDC retention and completeness over the certified interval.
- preservation of same-coordinate order and multiplicity where relevant.
- honesty of chunk-read values and absences at their recorded coordinates.
- the primary-key interpretation and the assumption that chunk reads observe committed source state.

The theorem statements write the raw source-observation carrier as *raw*. The carrier is abstract. It stands for whatever concrete record a deployment has of what the source emitted, interpreted by the faithful-source-observation assumption. It is not a destination log, not a sink-state proof, and not a claim that a named connector has been verified.

The certificate checker can compare evidence fields to one another. It cannot prove, from the certificate alone, that the outside world produced faithful observations. This is why the accepted-certificate theorem takes faithful source observation as a premise, labeled as an External observation assumption.

8.8 Role of the Certificate Layer

Layer 2 needs a wellformed mathematical run, while a deployment exposes a certificate and evidence. Layer 3 connects the two. It states that accepted evidence materializes a coherent wellformed run under faithful source observation, after which the source-side virtual-cut theorem applies.

Practitioner takeaway. *A certificate names the scope, frontier, and clean prefix. Its evidence must materialize a coherent wellformed run. Acceptance checks this internal relationship, while faithful source observation remains an explicit assumption.*

Table 7: Checker-substrate obligations S1–S4 for the accepted-certificate intermediate theorem.

Checker-substrate obligation	
S1	Accepted inputs are wellformed checker inputs that are structurally valid and parseable.
S2	Accepted checker inputs materialize an abstract DBLog run.
S3	The materialized run’s accessors (scope, frontier, and clean prefix) agree with the certificate.
S4	A faithful accepted materialization is a wellformed run against the source history.

9 Accepted-Certificate Soundness (Layer 3)

Layer 3 gives the certificate-side theorem. It has two statements: an intermediate bridge theorem and the composed virtual-cut theorem.

9.1 Intermediate Theorem

The intermediate theorem is machine-checked under the *checker substrate*, a small set of primitive obligations the abstract certificate checker must meet for acceptance to carry meaning. Table 7 lists them. They are stated as explicit assumptions. We do not prove that a concrete parser, wire format, or production verifier discharges them.

Obligations S1–S4 are the premise structure of the intermediate theorem. Each is a property that the checker design must satisfy before an *Accept* result can license the theorem’s source-side conclusion. Table 7 carries no fifth row because *Unsupported-evidence* reporting is not part of that premise list. It is a separate checker-design commitment, stated once outside the table.

Unsupported reporting. A checker meeting S1–S4 must also not accept evidence it cannot interpret as if it had been checked. Uninterpretable evidence is reported *Unsupported* rather than *Accept*. **Status:** Deployment obligation on the concrete checker, not a Layer 3 premise.

Under obligations S1–S4 the intermediate theorem is:

$$\begin{aligned} \text{verify}(C, E) &= \text{Accept} \wedge \text{FSO}(E, b_0, H, \text{raw}) \\ &\implies \exists R. \text{WF}(b_0, R, H) \wedge \text{Coh}(C, E, R). \end{aligned}$$

In words, accepted evidence, when interpreted through faithful source observation, witnesses a wellformed abstract DBLog run whose accessors agree with the certificate. Obligations S1–S3 carry the materialization and accessor agreement, and S4 carries wellformedness. The *Unsupported-reporting* commitment stated after the table is what keeps uninterpretable evidence from being accepted as if it had been checked. It is operational, not part of the premise list.

9.2 Composed Theorem

The main certificate theorem is:

$$\begin{aligned} \text{verify}(C, E) &= \text{Accept} \wedge \text{FSO}(E, b_0, H, \text{raw}) \\ &\implies \text{VC}(C, H). \end{aligned}$$

Expanded:

$$\begin{aligned} \text{verify}(C, E) &= \text{Accept} \wedge \text{FSO}(E, b_0, H, \text{raw}) \\ &\implies \text{apply}(\text{prefix}(C)) \upharpoonright \text{scope}(C) \\ &= \text{src}(b_0, H, \text{frontier}(C)) \upharpoonright \text{scope}(C). \end{aligned}$$

Reader-facing theorem summary:

- **Status:** Machine-checked under checker-substrate obligations S1–S4 (Table 7) and faithful source observation.
- **Claim:** an accepted certificate is a virtual-cut certificate when its evidence is interpreted through faithful source observation.
- **Premises:** checker-substrate obligations S1 (input wellformedness), S2 (materialization), S3 (accessor agreement), S4 (faithful materialization is wellformed), and faithful source observation of the evidence.
- **Boundary:** acceptance does not prove a concrete parser, wire format, production verifier, named product, whole-table claim, or sink application. The *Unsupported-reporting* commitment is a separate checker-design obligation, not part of the premise list.

This is the formal meaning of “an accepted DBLog certificate is a virtual-cut certificate.” The faithful-source-observation premise is part of the sentence, not a footnote. Acceptance alone is not a proof about the mathematical source history.

9.3 Proof Explanation

The proof composes the layers already introduced.

First, verifier acceptance and faithful source observation produce a witnessed run R :

$$\begin{aligned} &\text{WF}(b_0, R, H), \\ &\text{Coh}(C, E, R). \end{aligned}$$

Second, Layer 2 applies to that wellformed run:

$$\begin{aligned} &\text{apply}(\text{prefix}(R)) \upharpoonright \text{scope}(R) \\ &= \text{src}(b_0, H, \text{frontier}(R)) \upharpoonright \text{scope}(R). \end{aligned}$$

Third, coherence substitutes certificate accessors for run accessors:

$$\begin{aligned} \text{scope}(C) &= \text{scope}(R), \\ \text{frontier}(C) &= \text{frontier}(R), \\ \text{prefix}(C) &= \text{prefix}(R). \end{aligned}$$

After substitution, the Layer 2 equality becomes:

$$\begin{aligned} & \text{apply}(\text{prefix}(C)) \upharpoonright \text{scope}(C) \\ &= \text{src}(b_0, H, \text{frontier}(C)) \upharpoonright \text{scope}(C). \end{aligned}$$

Fourth, Section 6’s definition folds that equality into $\text{VC}(C, H)$.

The proof uses no sink, whole-table, or named-product premise. Its conclusion is source-side and restricted to $\text{scope}(C)$. The checker-substrate obligations and faithful source observation remain premises.

9.4 Witnesses and Boundary Cases

The formal development includes the following witness and boundary checks:

- there is a nonempty accepted-certificate witness with a non-base frontier, a chunk read, a value-changing replay event, and faithful $\text{raw}/b_0/H$ evidence.
- accepted-without-witness shapes are rejected by the checker-substrate obligations.
- wrong certificate/evidence binding is rejected.
- unknown source-side evidence is `Unsupported` rather than silently accepted.
- mismatched certificate/run accessors fail coherence.
- unfaithful raw , wrong b_0 , and wrong H fail faithful source observation.
- empty scope is a trivial boundary, not the positive witness.

These cases protect the main trust boundary. The theorem does not say that the certificate eliminates trust. It says that, under faithful source observation and explicit checker obligations, acceptance gives a coherent wellformed run. The source-side theorem then yields the certified virtual cut.

Practitioner takeaway. *A consumer can check a finite certificate against its evidence bundle. Under the stated checker obligations, acceptance yields a virtual cut only when the captured observations are faithful to the real source.*

10 Whole-Table Specialization (Layer 4)

Layers 2 and 3 prove equality on $\text{scope}(C)$. A full backfill often intends that scope to cover a whole table or shard. Such a claim needs additional evidence that the scope includes every key whose state can matter between an anchor boundary and the certified frontier. The replay must also avoid writing keys outside that scope.

The whole-table specialization is therefore a derived result, not a new meaning of certificate acceptance. Acceptance gives the scoped virtual cut. The additional side condition tells us when the restriction to $\text{scope}(C)$ can be dropped.

10.1 Anchor Boundary and Anchor Domain

An anchor boundary b is a source coordinate used to name the table population the whole-table claim intends to cover. It is not a physical snapshot read. Chunks may be read after b . Stale chunk observations can be repaired by source-log evidence and replayed to the frontier.

For a certificate C , the anchor must not be later than the certified frontier:

$$b \leq \text{frontier}(C).$$

The anchor domain is the set of keys present in the source at that boundary:

$$\text{anchorDom}(H, b) = \{ k \mid \text{src}(b_0, H, b)(k) \text{ is present} \}.$$

The anchor domain inherits its dependence on the base state b_0 from src . As with src , b_0 is a single base state fixed throughout the paper. We keep it implicit in $\text{anchorDom}(H, b)$, and likewise in the whole-table claim scope $\text{WTS}(C, H, b)$ below, since it never varies. A key present at the anchor that no source event at or before b ever writes owes its anchor-domain membership entirely to b_0 , the one place the base state does observable work in a whole-table claim.

Operationally, a whole-table deployment may use a maximum-primary-key or equivalent end-of-table witness to help show that the claimed scope covers this anchor domain. Such a witness does not replace WTS . It must be tied to the anchor boundary b , or otherwise justify $\text{anchorDom}(H, b) \subseteq \text{scope}(C)$. It also does not discharge the post-anchor touch condition below, including for keys beyond an anchor-time maximum.

Keys absent at b can still become relevant later. If the source inserts, updates, or deletes a key after b and at or before the certified frontier, the whole-table claim must account for that key too.

$\text{touched}(H, b, f)$ is the set of keys k for which H has an insert, update, or delete at some source coordinate c with $b < c \leq f$, and the conservative interval table scope is

$$\begin{aligned} & \text{tblScope}(H, b, f) \\ &= \text{anchorDom}(H, b) \cup \text{touched}(H, b, f). \end{aligned}$$

This is a conservative event-touch scope. It is not a minimal semantic-difference set. Even a delete of an already absent key is still a touch that the whole-table side condition can require the claim scope to cover.

10.2 A Small Anchor Example

Suppose the source contains keys 100 and 200 at anchor boundary b . Between b and frontier f , key 300 is inserted and key 200 is deleted.

$$\begin{aligned}
\text{anchorDom}(H, b) &= \{100, 200\}, \\
\text{touched}(H, b, f) &= \{200, 300\}, \\
\text{tblScope}(H, b, f) &= \{100, 200, 300\}.
\end{aligned}$$

A whole-table claim for this interval must include all three keys in $\text{scope}(C)$. Key 100 is present at the anchor. Key 200 is present at the anchor and then deleted. Key 300 is absent at the anchor but inserted before the frontier. Omitting any of them would make the whole-table statement stronger than the evidence supports.

There is also a replay-side condition. If the certified clean prefix writes a bystander key 999 outside $\text{scope}(C)$, then scoped equality over $\{100, 200, 300\}$ may still be true, but whole-table equality is not justified. The replay has changed something outside the claimed table scope.

10.3 Whole-Table Claim Scope

We write $\text{WTS}(C, H, b)$ for the whole-table claim-scope side condition. It has four conjuncts:

$$\begin{aligned}
\text{WTS}(C, H, b) &\iff b \leq \text{frontier}(C) \\
&\wedge \text{anchorDom}(H, b) \subseteq \text{scope}(C) \\
&\wedge \text{touched}(H, b, \text{frontier}(C)) \subseteq \text{scope}(C) \\
&\wedge \text{keys}(\text{prefix}(C)) \subseteq \text{scope}(C).
\end{aligned}$$

The first conjunct fixes the ordering of the anchor and frontier. The next two say that $\text{scope}(C)$ covers every key present at the anchor and every key touched after the anchor through the frontier. The final conjunct says that the certified replay prefix does not write unclaimed keys. Figure 5 pictures the four conjuncts as overlapping key regions (anchor-domain, touched-between, replay-keys) over the source-coordinate axis.

This predicate is explicit because whole-table correctness is not automatic. Certificate acceptance alone gives a scoped statement. The whole-table side condition is the extra bridge from “correct on the certified claim scope” to “correct on every table key”.

10.4 Whole-Table Theorem

Under certificate acceptance, faithful source observation, and the whole-table claim-scope side condition, the certified clean prefix reaches the source state at the certified frontier on the whole table:

$$\begin{aligned}
\text{verify}(C, E) &= \text{Accept} \wedge \text{FSO}(E, b_0, H, \text{raw}) \\
&\wedge \text{WTS}(C, H, b) \\
&\implies \text{apply}(\text{prefix}(C)) = \text{src}(b_0, H, \text{frontier}(C)).
\end{aligned}$$

Reader-facing theorem summary:

- **Status:** Machine-checked under checker-substrate obligations S1–S4 (Table 7), faithful source observation, and $\text{WTS}(C, H, b)$.
- **Claim:** the scoped certificate theorem lifts to whole-table source-state equality when the whole-table claim-scope side conditions cover boundary order, anchor-domain keys, post-anchor touches, and replay keys.
- **Premises:** certificate acceptance, faithful source observation, $\text{WTS}(C, H, b)$, and checker-substrate obligations S1–S4 (Table 7).
- **Boundary:** this remains source-side. It does not prove sink convergence, exactly-once delivery, or arbitrary destination correctness.

This is a source-side theorem. The certificate checker and faithful-source-observation assumption supply the certified scoped virtual cut. The whole-table side condition supplies the coverage needed to remove the scope restriction.

10.5 Proof Outline

Layer 3 gives the scoped equality:

$$\begin{aligned}
&\text{apply}(\text{prefix}(C)) \upharpoonright \text{scope}(C) \\
&= \text{src}(b_0, H, \text{frontier}(C)) \upharpoonright \text{scope}(C).
\end{aligned}$$

For keys inside $\text{scope}(C)$, that is already the desired equality. For keys outside $\text{scope}(C)$, the whole-table side condition does two jobs. The $\text{WTS}(C, H, b)$ inclusions

$$\begin{aligned}
&\text{anchorDom}(H, b) \subseteq \text{scope}(C), \\
&\text{touched}(H, b, \text{frontier}(C)) \subseteq \text{scope}(C), \\
&\text{keys}(\text{prefix}(C)) \subseteq \text{scope}(C)
\end{aligned}$$

contrapose to $k \notin \text{scope}(C) \implies k \notin \text{anchorDom}(H, b)$, $k \notin \text{scope}(C) \implies k \notin \text{touched}(H, b, \text{frontier}(C))$, and $k \notin \text{scope}(C) \implies k \notin \text{keys}(\text{prefix}(C))$, which power the two cases. First, the b_0 -aware anchor-domain definition makes absence outside the anchor domain mean absence at the anchor boundary, including keys from the base state. Together with $b \leq \text{frontier}(C)$ and no post-anchor touch through the frontier, this implies the key is absent from the source at the frontier. Second, because the key is outside $\text{keys}(\text{prefix}(C))$, the clean prefix does not write it. Replay and source are therefore both absent outside $\text{scope}(C)$. Combining the inside-scope and outside-scope cases gives unrestricted equality.

10.6 Scope of the Result

The specialization does not change the paper’s trust boundaries. It still depends on faithful observation of the source-side evidence. It does not prove that a concrete parser or production verifier implementation is correct. It does not prove sink-state convergence, exactly-once delivery, or arbitrary destination correctness. It also does not say that a virtual cut is a physical cut: the anchor is a source-coordinate boundary used for

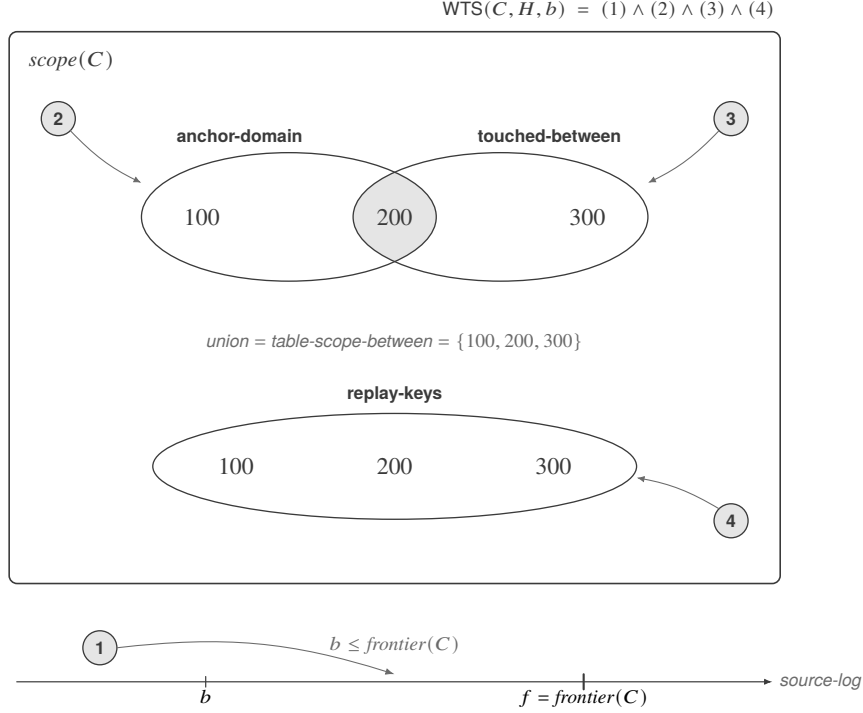


Figure 5: Anchor domain and whole-table claim scope. The four numbered conjuncts of $WTS(C, H, b)$ are annotated on the diagram. The base state b_0 , fixed by context, is implicit in WTS and $anchorDom$. (1) The anchor b precedes the frontier on the source-log axis: $b \leq frontier(C)$. (2) $anchorDom(H, b) \subseteq scope(C)$. (3) $touched(H, b, frontier(C)) \subseteq scope(C)$. (4) $keys(prefix(C)) \subseteq scope(C)$. Concrete keys follow the worked example of Section 10.2: 100 sits in anchor-domain only, 200 in the overlap, 300 in touched-between only. Replay-keys are $\{100, 200, 300\}$.

reasoning about table scope, not a single request-wide read timestamp.

Practitioner takeaway. A whole-table result requires coverage of every key present at the anchor and every key touched after the anchor and through the frontier. The certified replay must also write nothing outside that scope. Certificate acceptance alone gives only the scoped result.

11 Source-Side Continuation and Restriction of Virtual Cuts

The results so far fix replay equality at one frontier. After the chunk loop closes, DBLog continues consuming source-log events. We now show how the established equality advances with that stream.

Continuation across frontiers appends the source changes committed after frontier f and through a later frontier f' . Restriction shows that a virtual cut on scope K remains valid on every smaller scope. Both results are source-side and machine-checked. They contain no destination, replica, or delivery term. Disjoint composition, verification of extended certificates, and normalization of overlapping or retried chunks are not proved here.

Both proofs proceed per key. Continuation uses the Layer 0 fact that the source state at f' is determined by the state at f together with the events in $(f, f']$. Restriction follows directly from equality on each key in the original scope.

11.1 Continuation Segments

For a source history H , a key scope K , and source frontiers $f \leq f'$, the continuation segment for the half-open interval $(f, f']$ on K is obtained by filtering H . We keep every source event whose coordinate c satisfies $f < c \leq f'$ and whose key lies in K . The events remain in source order and are read as CDC replay events. We write $ContSeg(H, K, f, f', \delta)$ when δ is exactly this list, with source order and multiplicity preserved.

By construction, $ContSeg(H, K, f, f', \delta)$ entails $f \leq f'$. The predicate is false whenever $f > f'$, independent of δ . Theorems whose premises include $ContSeg(H, K, f, f', \delta)$ therefore implicitly require $f \leq f'$. The order is not restated in their displayed formulas.

The interval is half-open because the coordinate convention is inclusive (Section 5.1): $src(b_0, H, f)$ already reflects every event at coordinate f , so a continuation starts strictly after f and runs through f' inclusive.

Because the segment is defined by filtering the source history, a list that reorders, drops, or duplicates a source event does not satisfy $\text{ContSeg}(H, K, f, f', \delta)$. A real CDC feed must deliver this exact segment. Its faithfulness is an **External observation assumption**, and retaining the source-log interval $(f, f']$ is a **Deployment obligation**.

The segment is filtered to the scope K . It is the change data for the keys of K , not the source's entire change stream. A deployment whose CDC feed carries all keys must project that stream onto K to obtain the continuation segment for a K -scoped cut. For a whole-table continuation the scope is the universal key set, and the whole-table continuation result below is the form to use. The definition is exact. A segment that drops, reorders, or duplicates an *in-scope* event is not $\text{ContSeg}(H, K, f, f', \delta)$ and the continuation does not hold. A segment padded with an *out-of-scope* event also fails ContSeg , although the scoped equality itself constrains only the keys of K .

11.2 Continuation Across Frontiers

The primary theorem of this section continues a virtual cut to a later frontier. Let H be a wellformed source history. If σ is a virtual cut at frontier f on scope K , and δ is the continuation segment for $(f, f']$ on K , then σ followed by δ (sequence concatenation, written $\sigma @ \delta$) is a virtual cut at the later frontier f' on the same scope:

$$\begin{aligned} & H \text{ wellformed} \wedge \text{VCS}(\sigma, K, f, H) \\ & \wedge \text{ContSeg}(H, K, f, f', \delta) \\ & \implies \text{VCS}(\sigma @ \delta, K, f', H). \end{aligned}$$

Reader-facing theorem summary:

- **Status:** Machine-checked under the displayed premises.
- **Claim:** appending the faithful continuation segment for $(f, f']$ carries a virtual cut at f to a virtual cut at f' on the same scope.
- **Premises:** H is a wellformed source history, σ is a virtual cut at f on K , and δ is exactly the continuation segment for $(f, f']$ on K .
- **Boundary:** the conclusion is a replay-against-source equality on K . It is not a whole-table claim, not a statement that a verifier accepts an extended certificate, and not a destination-apply claim.

The wellformedness premise is needed because the proof uses the source history's coordinate order to know that the events in $(f, f']$ are exactly the ones between the two frontiers. Without it the interval could interleave with events outside it.

Proof idea. The argument is pointwise on a key $k \in K$. Replay of a sequence at k is fixed by the latest replay event for k . If the continuation segment δ carries an event for k , then the latest k -event of $\sigma @ \delta$ lies in δ and is the change-data reading of the latest source event for k in $(f, f']$. A Layer 0 source-state interval lemma equates that event's effect with

$\text{src}(b_0, H, f')$ at k . If δ carries no event for k , replay of $\sigma @ \delta$ at k equals replay of σ at k , which the premise equates with $\text{src}(b_0, H, f)$ at k . The same interval lemma equates that with $\text{src}(b_0, H, f')$ at k , because no event for k in the interval means the source state for k does not change across it. The interval lemma (that src at f' is determined by src at f together with the source events in $(f, f']$) is the one new Layer 0 fact the continuation needs. It is proved from the source-coordinate order.

Whole-table continuation. The whole-table specialization (Section 10) delivers, under its claim-scope side conditions, the *unrestricted* equality $\text{apply}(\sigma) = \text{src}(b_0, H, f)$. Continuation specializes to that case. For a wellformed H , if $\text{apply}(\sigma) = \text{src}(b_0, H, f)$ holds on every key and δ is the continuation segment for $(f, f']$ on every key, then $\text{apply}(\sigma @ \delta) = \text{src}(b_0, H, f')$. This is the all-keys instance of the theorem above, machine-checked under the same premises. It does not let a *scoped* continuation become whole-table without the all-keys premise. A key the source inserts outside K within $(f, f']$ changes the table while leaving the scoped equality on K intact. Whole-table continuation requires the continuation segment for every key, just as whole-table specialization requires its claim-scope side condition.

11.3 Restriction to a Sub-Scope

A virtual cut also restricts downward along the scope. If σ is a virtual cut at frontier f on scope K , then it is a virtual cut at f on every sub-scope $K' \subseteq K$:

$$\text{VCS}(\sigma, K, f, H) \wedge K' \subseteq K \implies \text{VCS}(\sigma, K', f, H).$$

This is machine-checked. It holds for any $K' \subseteq K$, and the $K' \subseteq K$ premise is exactly what it needs. Restriction is one-directional. *Widening* a scope is unsound. For a key outside K the cut certifies nothing about that key's state at f , so there is no sound lemma carrying a cut from K to a larger scope.

The same fact holds through a certificate's accessors. If C is a certificate with $\text{VC}(C, H)$ and $K' \subseteq \text{scope}(C)$, then the certified clean prefix is a virtual cut at $\text{frontier}(C)$ on the narrowed scope K' :

$$\begin{aligned} & \text{VC}(C, H) \wedge K' \subseteq \text{scope}(C) \\ & \implies \text{VCS}(\text{prefix}(C), K', \text{frontier}(C), H). \end{aligned}$$

This restricts the *equality* a certificate already witnesses to a smaller scope. It is not a claim that a verifier accepts a restricted certificate. Producing and accepting a concrete certificate for K' is an evidence-and-verifier obligation, not a consequence of the algebra.

11.4 Continuation Over an Accepted Certificate

Continuation composes with the certificate layer. Layer 3 (Section 9) proves that an accepted certificate, under faithful source observation, witnesses a virtual cut at $\text{frontier}(C)$ on $\text{scope}(C)$. Appending a continuation segment carries that cut forward. Let H be a wellformed source history. If a certificate C is accepted under faithful source observation, and δ is the continuation segment for $(\text{frontier}(C), f']$ on $\text{scope}(C)$, then the certified clean prefix followed by δ is a virtual cut at f' on $\text{scope}(C)$:

$$\begin{aligned} \text{verify}(C, E) &= \text{Accept} \wedge \text{FSO}(E, b_0, H, \text{raw}) \\ &\wedge H \text{ wellformed} \\ &\wedge \text{ContSeg}(H, \text{scope}(C), \text{frontier}(C), f', \delta) \\ &\implies \text{VCS}(\text{prefix}(C) @ \delta, \text{scope}(C), f', H). \end{aligned}$$

Reader-facing theorem summary:

- **Status:** Machine-checked under checker-substrate obligations S1–S4 (Table 7), faithful source observation, source-history wellformedness, and exact continuation on $(\text{frontier}(C), f']$.
- **Claim:** an accepted certificate’s virtual cut continues, by appending a faithful continuation segment, to a virtual cut at a later frontier.
- **Premises:** certificate acceptance under checker-substrate obligations S1–S4 (Table 7) and faithful source observation, as in Layer 3, a wellformed source history, and a continuation segment that is exactly the source’s change data on $(\text{frontier}(C), f']$.
- **Boundary:** the conclusion is a replay equality on the *sequence prefix*(C) @ δ . It is not a claim that the verifier accepts an “extended certificate.” Acceptance for the longer prefix remains an evidence-and-verifier obligation.

The continuation segment δ is itself an external observation of the source. Its faithfulness to H is a premise discharged at deployment, exactly as the certificate’s own evidence is. The theorem composes the Layer 3 certificate theorem with continuation across frontiers. The composition leaves the trust assumptions for the certificate evidence and the continuation segment unchanged.

11.5 Ongoing Operation as a Sequence of Cuts

Continuation segments chain. For a wellformed source history, the continuation segment for $(f, f']$ followed by the continuation segment for $(f', f'']$ is the continuation segment for $(f, f'']$. Wellformedness is again required. Only a non-decreasing source history guarantees that the two coordinate ranges meet at f' without interleaving. A finite prefix of ongoing change data is a fold of continuation segments, and a virtual cut can be advanced one segment at a time or many at once with the same result.

DBLog’s replay-equivalence claim can therefore advance across frontiers. Each later cut consists of the original cut followed by the source changes committed in between.

Re-chunking and re-running. The same algebra explains what a fresh DBLog run produces. A virtual cut at (K, f) is determined by the source state $\text{src}(b_0, H, f)$ alone. Any two replay sequences that are virtual cuts at (K, f) agree on K , because both equal that one source state there. So a fresh *well-formed* run (or a fresh accepted certificate, under the Layer 2 and Layer 3 theorems already proved) regenerates the same absolute source-side virtual cut on its scope, independent of any earlier cut and independent of any replica’s state. Re-chunking is therefore an absolute, source-determined reconstruction of the per-key state (the regenerated clean prefix need not be the same event sequence, but replaying it reaches the same source state), not a differential repair against a destination. Whether a destination then applies the regenerated prefix is a separate, sink-side question, which Section 12 leaves outside the current claims.

11.6 Scope of the Results

Each conclusion is a source-side *apply-against-src* equality. The continuation segment must be the source’s real change data on the interval, which is an External observation assumption with a Deployment obligation to retain that interval. These results prove no whole-table claim beyond the explicit all-keys instance, no acceptance of an extended or restricted certificate, and no destination-state convergence. Continuation and restriction are machine-checked. The remaining certificate-algebra program is not proved here.

Practitioner takeaway. *Appending the source changes in $(f, f']$ carries a virtual cut at f to a cut at f' . A cut can also be restricted to a smaller scope. Re-running a wellformed backfill reconstructs the same source state on its scope, independent of the current destination state.*

12 Limits on Sink-State Convergence

Layers 2 through 4 are source-side. They relate the certified replay prefix to the source at the certified frontier. They do not establish the state of a downstream destination.

Three destination properties must be distinguished. An *ACK* records receipt. A *durable audit log* persists what was received. *Applied state* is the state reached by the destination’s tables.

12.1 Receipt ACK

A destination ACK says that some material was received. It does not say that the destination applied that material, applied it in order, applied it idempotently, applied it only once, or reached the source state at the certified frontier. Receipt could

support a future stream-correctness result, but it does not establish sink-state correctness.

12.2 Durable Audit-Log Persistence

A destination may durably record the certified clean prefix together with a run id, scope, frontier, or digest. This gives an auditable record of what arrived. Durable receipt is still stream-level evidence, not state-level evidence. A log can contain the right events while the destination table has not applied them or has applied them under different semantics.

12.3 Applied-State Evidence

To prove destination state convergence, a future layer would need a sink model with ordering, idempotence, fencing, visible apply barriers, and a way to observe that the destination state after the barrier corresponds to applying the certified clean prefix. This extension remains future work.

Even that future theorem would be scoped. Using *sinkState* as a placeholder for a future sink-state accessor, the target statement has the following form:

$$\textit{sinkState} \upharpoonright \textit{scope}(C) = \textit{src}(b_0, H, \textit{frontier}(C)) \upharpoonright \textit{scope}(C).$$

Whole-table sink convergence would additionally need the whole-table side conditions from Section 10. There is no path from “DBLog accepted a certificate” to “the whole destination database is correct” without both the source-side certificate theorem and a separate sink contract.

Practitioner takeaway. *A virtual cut relates replay to the source. An ACK establishes receipt, and an audit log establishes persistence. A destination-state guarantee additionally needs a fenced, idempotent apply contract and visible barrier evidence.*

13 Related Work

The closest work comes from change-data-capture platforms, distributed cuts and frontiers, stream-processing watermarks, database recovery, and formal certificates. We compare these areas by concept and use named systems as deployment context.

13.1 DBLog Mechanism Lineage and Production Adopters

The DBLog mechanism was introduced in the December 2019 Netflix Tech Blog post [4] and described operationally in the 2020 DBLog paper [1]. Online primary-key chunk reads are bracketed by source watermarks, chunk rows become refresh events, ordinary CDC events fill the log gaps, and stale refresh rows are suppressed when newer log evidence dominates

them. We formalize the result as a certified virtual cut. The model uses a canonical, non-overlapping chunk partition. The original mechanism’s retried or overlapping chunk reads must first be normalized into that canonical form. That normalizer is not developed here (Section 14.3).

Debezium [2] and Apache Flink CDC [3] include DBLog-style mechanisms in open-source CDC deployments. Debezium provides incremental snapshot modes, including signaling-table watermarks and read-only variants based on observed source coordinates such as MySQL GTIDs [5], MariaDB GTIDs [6], or PostgreSQL log sequence numbers (LSNs) [7]. Apache Flink CDC combines Debezium-style connectors with Flink’s checkpointed streaming runtime [8], including parallel chunk readers.

The formal result does not transfer to these systems automatically. A deployment must still establish source-log retention, watermark or coordinate semantics, frontier discipline, and faithful source observation.

13.2 Cuts, Frontiers, Watermarks, and Checkpoints

The term “virtual cut” follows the distributed-systems language of logical time and consistent cuts. Lamport’s logical clocks [9], Chandy-Lamport snapshots [10], Mattern’s virtual time [11], Fidge’s vector clocks [12], and Pu’s on-the-fly consistent reading [13] are the nearest theoretical neighbors.

A DBLog virtual cut is not a global consistent cut over all processes. It is an extensional replay-equality statement. Replaying a certified prefix yields the source state at a frontier on a key scope. The ordering substrate is a source-history coordinate, not a vector-clock model of inter-process channels. Each cut is point-in-time, fixed to one frontier, but the cuts chain. The source-side continuation result (Section 11) advances a cut from one frontier to a later one, so a DBLog run corresponds to a sequence of source-side virtual cuts rather than a single one.

Stream-processing systems use adjacent terms. The Dataflow model [14] popularized event time and watermarks. Naiad [15] uses frontiers as progress markers. Flink’s asynchronous snapshots [8] checkpoint operator state for recovery. DBLog’s watermarks are source evidence around chunk reads, and *frontier(C)* is the source-history boundary for replay equality. It is not a stream-processing checkpoint of operator state, and it is not an exactly-once processing claim.

13.3 CDC Infrastructure and Source-Log Semantics

Durable transport and coordination systems such as LinkedIn Databus [16], Facebook Wormhole [17], Kafka [18], Kafka Connect JDBC [19], and ZooKeeper [20] show the operating environment for CDC backfills. They do not export the certificate object defined here.

Source-database documentation for MySQL binary logs and GTIDs [5], PostgreSQL logical decoding [7], PostgreSQL logical replication [21], and MariaDB GTIDs [6] identifies the source-log ordering and row-event semantics that a deployment must establish for faithful source observation.

Database-native and cloud CDC systems include PostgreSQL logical replication [21], CockroachDB changefeeds [22], AWS DMS [23], GCP Datastream [24], Oracle GoldenGate [25], Striim [26], Materialize’s PostgreSQL ingestion [27], Databricks Lakeflow [28], and Microsoft SQL Server CDC [29]. These systems belong to the broader “initial load plus change stream” family. Our result adds a finite replay-equivalence certificate for the DBLog form of this pattern, under explicit source-observation assumptions.

CockroachDB resolved timestamps provide a production analogue of a source frontier. PostgreSQL logical replication illustrates the initial-copy-plus-ordered-change-stream pattern. Neither is treated here as exporting the finite, scoped evidence required by a virtual-cut certificate.

13.4 Recovery, Snapshots, Isolation, and Live Backfill

A database snapshot may refer to a recovery snapshot, a multi-version concurrency control (MVCC) read view, or point-in-time recovery (PITR). ARIES [30] and PostgreSQL PITR [31] reconstruct database state from checkpoints and logs. CALC [32] shows that database systems can construct logically consistent checkpoints without conventional synchronous physical checkpointing. Our contribution is different. It provides a scoped, source-side replay-equivalence certificate for a DBLog backfill, whose clean replay prefix equals source state at a named frontier and can be advanced to later frontiers. Berenson et al. [33], Adya et al. [34], and Oracle Flashback [35] belong to the transaction-isolation and as-of-read tradition.

DBLog is not a recovery algorithm, an online backup mechanism, an MVCC snapshot, or a snapshot-isolation transaction. It does not assert that all chunk reads happened at one source timestamp. It asserts replay equality at a chosen frontier on a chosen scope.

Online schema migration tools such as gh-ost [36], F1 online schema change [37], and lazy schema migration systems [38] share the operational pressure of copying or rewriting data while writes continue. Their overlap with DBLog is live backfill under concurrent writes. The formal target differs. We certify source-side replay equality for a DBLog-style backfill and its frontier advancement, not schema-state transitions or table swaps.

13.5 Certificate Discipline

The certificate layer follows the general idea of checkable evidence under explicit assumptions. Refinement mappings

[39], proof-carrying code [40], translation validation [41], version certificate recovery [42], and VerIso [43] all help position the idea of a finite object that witnesses correctness relative to a model. Large mechanized artifacts such as seL4 [44], CompCert [45], TLA+ [46], Verdi [47], IronFleet [48], and Alloy [49] are cited for discipline and comparison, not as theorem sources.

A DBLog certificate is not self-authenticating proof-carrying code, not a sink refinement map, and not an isolation-level conformance certificate. It is checkable evidence that, under faithful source observation, witnesses a wellformed DBLog run and therefore a virtual cut.

13.6 Scope of the Comparison

“Snapshot-equivalent replay” is intended literally but narrowly. A DBLog virtual cut is not a physical snapshot read, not a database snapshot-isolation read view, not PITR, not a stream-processing checkpoint, not a product replication guarantee, and not sink-state correctness. The result is a snapshot-equivalent replay certificate that does not require one physical snapshot read.

Debezium, Apache Flink CDC, MySQL, PostgreSQL, MariaDB, and the other named systems provide mechanism or deployment context. Applying the theorem to a concrete deployment requires its source-side assumptions, evidence semantics, and checker obligations to be established.

Practitioner takeaway. *The named CDC products show where DBLog-style mechanisms are used. A product satisfies the theorem only after its source-log, coordinate, observation, and checker assumptions have been matched to the formal model.*

14 Future Work

Section 11 proves continuation across frontiers and restriction to a sub-scope. The following extensions are not proved here.

14.1 Production Instantiation Paths

Debezium read-only mode would replace source watermark writes with observed source-coordinate witnesses, such as MySQL executed-GTID-set advance, PostgreSQL LSN advance, or related source-log observations. A future theorem would need a refinement showing that such observed-coordinate brackets provide the same clean-prefix obligations as written watermarks, under an explicit coordinate advancement assumption.

Apache Flink CDC parallel chunk execution would require a parallel chunk-plan model, per-worker recovery assumptions, and a theorem connecting checkpointed parallel execution to the same abstract wellformed run used here. The certificate

would need enough evidence to recover the single clean prefix consumed by the proof.

SELECT-at-LSN and source-positioned witness modes would replace low/high watermark intervals with a point witness at the source coordinate where a chunk read’s row image is known to hold. These modes are attractive because they can avoid source writes, but they need new source-side assumptions and a new verification model for the point witness.

14.2 Sink Extensions

Destination-side results require a separate sink model.

ACK-only stream correctness would study receipt of the certified stream prefix. An ACK can support a theorem about receipt, but not about applied destination state. Such a proof would need a sink-receipt model and a stream-prefix equality theorem.

Strong sink barrier convergence would establish applied destination state. It needs a separate fenced and idempotent sink contract, visible apply-barrier evidence, and a proof that the destination state after the barrier realizes the certified clean prefix. It is not a consequence of the accepted-certificate theorem or the whole-table specialization by itself.

14.3 Certificate Algebra and Model Extensions

We prove restriction to a sub-scope and continuation across later frontiers by faithful CDC segments. Disjoint composition, concrete extended-certificate verification, and raw overlap/retry normalization remain future work.

Real deployments also need a raw overlap/retry normalizer theorem. The current Layer 1 model uses a strict canonical chunk partition. A future normalizer would take overlapping or retried raw chunk reads and produce a canonical strict-partition certificate plus residual obligations, then prove that acceptance after normalization preserves the same virtual-cut claim.

Some notation and layer-boundary choices also remain open. A future paper might expose $prefix(C, E)$ instead of $prefix(C)$, use a relation form such as *certifiesPrefix*, or move overlap/retry concepts into Layer 1 instead of treating them through certificate materialization. Those changes would alter the public theorem interface and require an explicit revision.

15 Verification Methodology

Table 1 distinguishes machine-checked results from conditions supplied by a deployment or source connector. A result labeled **Machine-checked** is a theorem of the companion Isabelle/HOL development under the premises stated with that result. The paper gives the corresponding definition,

theorem statement, premises, and proof explanation in the body.

The formal development checks the source and replay model, the wellformed-run theorem, accepted-certificate soundness, the whole-table specialization, continuation, and restriction. It does not establish deployment facts such as source-log retention, coordinate ordering, frontier selection, or chunk scope. It also cannot determine whether a connector’s CDC records and chunk-row images faithfully describe the real source. These conditions are labeled **Deployment obligation** or **External observation assumption** where they are used.

The labels **Future work** and **Non-claim** identify results that are not part of the current theorem set. The first names an open extension. The second marks a property that the current result does not establish.

The paper and formal development serve different roles. The paper gives the argument in reader-facing form. The Isabelle development checks the formal definitions and proofs against one another. Neither substitutes for the source and deployment assumptions stated with the theorems.

16 Conclusion

In this paper, we formalized the correctness property behind DBLog’s live backfill mechanism. DBLog copies a table or a set of selected keys while the source continues accepting writes. It combines chunk reads with ordered CDC events without holding a long-lived table lock. Because different chunks may be read at different source-log coordinates, the collected rows do not form a physical snapshot.

We define the resulting object as a certified virtual cut. A virtual cut holds when applying a finite replay prefix produces the same per-key state as the source at a chosen frontier on a chosen scope. The proof reasons separately about each key. For each key, the latest relevant replay event is either the latest retained source event through the frontier or the responsible chunk’s refresh value or absence when no later source event supersedes it.

We prove this equality for every wellformed DBLog run. We then connect the run-level theorem to accepted certificates under faithful source observation and explicit checker assumptions. A whole-table result follows when the certificate also covers the required anchor and post-anchor keys. Finally, we prove that a virtual cut can be advanced by appending the source events committed between two frontiers, and that an existing cut can be restricted to a smaller scope. All of these results are machine-checked in Isabelle/HOL.

A certificate can expose missing or inconsistent evidence, but it cannot prove that a connector’s observations are faithful to the real database. Source-log retention, coordinate semantics, frontier selection, and chunk-read consistency remain deployment conditions. The results also do not establish

exactly-once delivery, applied destination state, correctness of a concrete verifier implementation, or correctness of a named product.

The contribution is therefore a source-side account of DBLog. The initial backfill constructs a replay that matches the source at one frontier, and continued CDC processing advances that equality as the frontier moves. This gives the combination of initial load and change capture a finite correctness object and a machine-checked statement of the conditions under which it holds.

Formal Development and Tools

The companion Isabelle/HOL development machine-checks the definitions and theorems under their stated assumptions. It does not discharge the *Deployment obligations* or the *External observation assumption*, which a deployment must establish separately. The formal development is publicly archived as the Zenodo deposit [50] (<https://doi.org/10.5281/zenodo.21732790>). The archived development declares no axioms. Every witness and counterexample fixture is a constructed instance. The only type not introduced as a datatype or record is a conservative typedef for source coordinates.

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